

Unit 6. Shrinkage, High Dimensions, KNN

1 Ridge Regression

ISLR §6.2.1.

Throughout §1 and §2, $\mathbf{Y} \in \mathbb{R}^n$ and $\mathbf{X}$ is $n \times k$ with columns $\mathbf{x}_1, \dots, \mathbf{x}_k$; no column of ones is included, the intercept being treated separately in §1.2. No rank condition is imposed on $\mathbf{X}$, and k may exceed n .

1.1 The estimator

Definition 6.1 (Ridge regression). For $\lambda \geq 0$ the **ridge** criterion is

$$Q_\lambda(\boldsymbol{\beta}) = |\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}|^2 + \lambda|\boldsymbol{\beta}|^2 = \sum_{i=1}^n \left(Y_i - \sum_{j=1}^k x_{ij}\beta_j \right)^2 + \lambda \sum_{j=1}^k \beta_j^2, \quad (1)$$

and a **ridge estimator** is a minimiser of Q_λ over $\mathbb{R}^k$.

Theorem 6.2 (Existence, uniqueness, closed form). For every $\lambda > 0$ the matrix $\mathbf{X}'\mathbf{X} + \lambda\mathbf{I}$ is positive definite, Q_λ has exactly one minimiser, and

$$\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}'\mathbf{Y}. \quad (2)$$

No condition on $\text{rank}(\mathbf{X})$ is needed.

Proof. For $\mathbf{c} \neq \mathbf{0}$, $\mathbf{c}'(\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})\mathbf{c} = |\mathbf{X}\mathbf{c}|^2 + \lambda|\mathbf{c}|^2 > 0$, so $\mathbf{X}'\mathbf{X} + \lambda\mathbf{I}$ is positive definite and in particular invertible.

Expanding (1),

$$Q_\lambda(\boldsymbol{\beta}) = |\mathbf{Y}|^2 - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{Y} + \boldsymbol{\beta}'(\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})\boldsymbol{\beta}. \quad (3)$$

Write $\mathbf{A} = \mathbf{X}'\mathbf{X} + \lambda\mathbf{I}$ and $\hat{\boldsymbol{\beta}} = \mathbf{A}^{-1}\mathbf{X}'\mathbf{Y}$. Completing the square,

$$Q_\lambda(\boldsymbol{\beta}) = |\mathbf{Y}|^2 - \hat{\boldsymbol{\beta}}'\mathbf{A}\hat{\boldsymbol{\beta}} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})'\mathbf{A}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}),$$

since $\boldsymbol{\beta}'\mathbf{A}\boldsymbol{\beta} - 2\boldsymbol{\beta}'\mathbf{A}\hat{\boldsymbol{\beta}} = (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})'\mathbf{A}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) - \hat{\boldsymbol{\beta}}'\mathbf{A}\hat{\boldsymbol{\beta}}$ and $\mathbf{A}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}$. The first two terms are free of $\boldsymbol{\beta}$ and the third is positive unless $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$. $\square$

Remark. At $\lambda = 0$ the criterion is least squares, and (2) reduces to the normal equations of Unit 2, which require $\text{rank}(\mathbf{X}) = k$ to be solvable uniquely. The term $\lambda \mathbf{I}$ removes that requirement: a rank-deficient or wider-than-tall design has a unique ridge estimator for every $\lambda > 0$.

Proposition 6.3 (Constrained form). Fix $s > 0$. If $\hat{\boldsymbol{\beta}}$ minimises $|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}|^2$ subject to $|\boldsymbol{\beta}|^2 \leq s$, then there is $\lambda \geq 0$ with $\hat{\boldsymbol{\beta}}$ minimising Q_λ . Conversely, if $\hat{\boldsymbol{\beta}}$ minimises Q_λ for some $\lambda \geq 0$, then $\hat{\boldsymbol{\beta}}$ minimises $|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}|^2$ subject to $|\boldsymbol{\beta}|^2 \leq |\hat{\boldsymbol{\beta}}|^2$.

Proof. Both halves are the case $P(\boldsymbol{\beta}) = |\boldsymbol{\beta}|^2$ of Theorem 6.17 below, whose proof uses nothing about the penalty beyond convexity and vanishing at the origin. $\square$

1.2 The intercept and the units of measurement

Proposition 6.4 (The intercept is not penalised). Let $\mathbf{X}$ have centred columns, $\mathbf{1}'\mathbf{x}_j = 0$ for each j , and consider minimising $|\mathbf{Y} - \beta_0\mathbf{1} - \mathbf{X}\boldsymbol{\beta}|^2 + \lambda|\boldsymbol{\beta}|^2$ over $(\beta_0, \boldsymbol{\beta})$. The minimiser has $\hat{\beta}_0 = \bar{Y}$, and $\hat{\boldsymbol{\beta}}$ is the ridge estimator (2) computed from the centred response $\mathbf{Y} - \bar{Y}\mathbf{1}$.

Proof. For fixed $\boldsymbol{\beta}$ the criterion is a quadratic in β_0 with minimiser $\hat{\beta}_0 = \mathbf{1}'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})/n = \bar{Y}$, the second term vanishing because every column of $\mathbf{X}$ is centred. Substituting, the criterion becomes $|(\mathbf{Y} - \bar{Y}\mathbf{1}) - \mathbf{X}\boldsymbol{\beta}|^2 + \lambda|\boldsymbol{\beta}|^2$, which is (1) with the centred response. $\square$

Proposition 6.5 (Least squares is scale equivariant, ridge is not). Let $\mathbf{D} = \text{diag}(d_1, \dots, d_k)$ with every $d_j > 0$, and replace $\mathbf{X}$ by $\mathbf{XD}$.

- (i) Least squares: if $\hat{\boldsymbol{\beta}}$ minimises $|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}|^2$ then $\mathbf{D}^{-1}\hat{\boldsymbol{\beta}}$ minimises $|\mathbf{Y} - \mathbf{XD}\boldsymbol{\beta}|^2$, and the two fitted vectors are equal.
- (ii) Ridge: $\hat{\boldsymbol{\beta}}_\lambda(\mathbf{XD}) = \mathbf{D}^{-1}\boldsymbol{\gamma}$, where $\boldsymbol{\gamma}$ minimises

$$|\mathbf{Y} - \mathbf{X}\boldsymbol{\gamma}|^2 + \lambda \sum_{j=1}^k \frac{\gamma_j^2}{d_j^2}. \quad (4)$$

Proof. Substitute $\boldsymbol{\gamma} = \mathbf{D}\boldsymbol{\beta}$, so that $\mathbf{XD}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\gamma}$ and $\boldsymbol{\beta} = \mathbf{D}^{-1}\boldsymbol{\gamma}$. The sum of squares depends on $\boldsymbol{\beta}$ only through $\boldsymbol{\gamma}$, which gives (i). The penalty is $|\boldsymbol{\beta}|^2 = |\mathbf{D}^{-1}\boldsymbol{\gamma}|^2 = \sum_j \gamma_j^2/d_j^2$, which gives (ii). $\square$

Remark. Changing the unit in which $\mathbf{x}_j$ is recorded changes the ridge fit, because it changes the weight λ/d_j^2 that the penalty places on that coordinate: a column recorded in smaller units has a larger d_j and is penalised less. Since a choice of unit carries no information about the response, the columns are standardised before penalising. The convention used throughout is

$$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{\hat{s}_j}, \quad \hat{s}_j^2 = \frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2, \quad (5)$$

so that each column has mean 0 and $|\tilde{\mathbf{x}}_j|^2 = n$. Divisor n rather than $n - 1$ is the convention `glmnet` uses internally, and matching it is what makes the comparisons of §4 exact.

1.3 Shrinkage of the principal components

Theorem 6.6 (Ridge through the singular value decomposition). Let $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}'$ be the singular value decomposition of the prerequisites, with $r = \text{rank}(\mathbf{X})$, $\mathbf{U}$ of size $n \times r$ and $\mathbf{V}$ of size $k \times r$

having orthonormal columns, and $\mathbf{D} = \text{diag}(d_1, \dots, d_r)$ with $d_1 \geq \dots \geq d_r > 0$. For $\lambda > 0$,

$$\hat{\boldsymbol{\beta}}_\lambda = \sum_{j=1}^r \frac{d_j}{d_j^2 + \lambda} (\mathbf{u}_j' \mathbf{Y}) \mathbf{v}_j, \quad (6)$$

$$\hat{\mathbf{Y}}_\lambda = \mathbf{X} \hat{\boldsymbol{\beta}}_\lambda = \sum_{j=1}^r \frac{d_j^2}{d_j^2 + \lambda} (\mathbf{u}_j' \mathbf{Y}) \mathbf{u}_j. \quad (7)$$

Proof. Write $\boldsymbol{\beta}^*$ for the right-hand side of (6), that is $\boldsymbol{\beta}^* = \mathbf{V}(\mathbf{D}^2 + \lambda \mathbf{I})^{-1} \mathbf{D} \mathbf{U}' \mathbf{Y}$. Using $\mathbf{X}' \mathbf{X} = \mathbf{V} \mathbf{D}^2 \mathbf{V}'$ and $\mathbf{V}' \mathbf{V} = \mathbf{I}_r$,

$$(\mathbf{X}' \mathbf{X} + \lambda \mathbf{I}) \boldsymbol{\beta}^* = \mathbf{V} \mathbf{D}^2 (\mathbf{D}^2 + \lambda \mathbf{I})^{-1} \mathbf{D} \mathbf{U}' \mathbf{Y} + \lambda \mathbf{V} (\mathbf{D}^2 + \lambda \mathbf{I})^{-1} \mathbf{D} \mathbf{U}' \mathbf{Y} = \mathbf{V} \mathbf{D} \mathbf{U}' \mathbf{Y} = \mathbf{X}' \mathbf{Y},$$

the two middle terms combining because $\mathbf{D}^2 + \lambda \mathbf{I}$ commutes with its own inverse. So $\boldsymbol{\beta}^*$ solves the normal equations, and by Theorem 6.2 it is the unique ridge estimator. Applying $\mathbf{X} = \mathbf{U} \mathbf{D} \mathbf{V}'$ to it gives $\mathbf{X} \boldsymbol{\beta}^* = \mathbf{U} \mathbf{D} (\mathbf{D}^2 + \lambda \mathbf{I})^{-1} \mathbf{D} \mathbf{U}' \mathbf{Y}$, which is (7). $\square$

Corollary 6.7 (Ridge shrinks the principal component coefficients). Let $\mathbf{X}$ be centred, so that by Unit 5 the loading vectors of its principal components are the $\mathbf{v}_j$, the score vectors are $\mathbf{z}_j = \mathbf{X}\mathbf{v}_j = d_j\mathbf{u}_j$, and the j -th sample variance is $d_j^2/(n-1)$. Write $\hat{\theta}_j = \mathbf{z}_j'\mathbf{Y}/|\mathbf{z}_j|^2$ for the coefficient of $\mathbf{z}_j$ in the least squares fit. Then

$$\hat{\mathbf{Y}}_\lambda = \sum_{j=1}^r \frac{d_j^2}{d_j^2 + \lambda} \hat{\theta}_j \mathbf{z}_j, \quad \hat{\mathbf{Y}}_0 = \sum_{j=1}^r \hat{\theta}_j \mathbf{z}_j. \quad (8)$$

Proof. The scores are orthogonal by Unit 5, so the least squares fit on $\mathbf{z}_1, \dots, \mathbf{z}_r$ is the sum of the separate projections, giving the second display and $\hat{\theta}_j = \mathbf{z}_j'\mathbf{Y}/d_j^2$. Substituting $\mathbf{u}_j = \mathbf{z}_j/d_j$ into (7) gives $\hat{\mathbf{Y}}_\lambda = \sum_j \{d_j^2/(d_j^2 + \lambda)\}(\mathbf{z}_j'\mathbf{Y}/d_j^2)\mathbf{z}_j$. $\square$

Remark. The factor $d_j^2/(d_j^2 + \lambda)$ lies in $(0, 1)$, increases in d_j and decreases in λ . Ridge therefore shrinks hardest along the directions in which the design varies least, which are exactly the directions in which the least squares coefficient is least well determined. Principal components regression makes the same distinction discretely, multiplying the first M coefficients by 1 and the rest by 0;

ridge replaces that step function by a smooth one.

Proposition 6.8 (Conditioning). If $\text{rank}(\mathbf{X}) = k$ then, in the notation of the prerequisites,

$$\kappa(\mathbf{X}'\mathbf{X} + \lambda\mathbf{I}) = \frac{d_1^2 + \lambda}{d_k^2 + \lambda} \leq \frac{d_1^2}{d_k^2} = \kappa(\mathbf{X}'\mathbf{X}), \quad (9)$$

with strict inequality when $\lambda > 0$ and $d_1 > d_k$.

Proof. $\mathbf{X}'\mathbf{X} + \lambda\mathbf{I} = \mathbf{V}(\mathbf{D}^2 + \lambda\mathbf{I})\mathbf{V}'$ is symmetric positive definite with eigenvalues $d_j^2 + \lambda$, so its condition number is the ratio of the extreme ones by the operator-norm proposition of the prerequisites. For $a > b > 0$ the map $t \mapsto (a + t)/(b + t)$ has derivative $(b - a)/(b + t)^2 < 0$, so it is strictly decreasing. $\square$

Remark. Collinearity was measured in Unit 4 by the variance inflation factor and in the prerequisites by $\kappa(\mathbf{X}'\mathbf{X})$. Ridge attacks it directly: adding λ to every eigenvalue leaves the large ones nearly unchanged and lifts the small ones away from zero.

1.4 Effective degrees of freedom

Definition 6.9 (Linear smoother). A fitting rule is a **linear smoother** if $\hat{\mathbf{Y}} = \mathbf{S}\mathbf{Y}$ for a matrix $\mathbf{S}$ determined by the design alone. Its **effective degrees of freedom** are $\text{tr}(\mathbf{S})$.

Proposition 6.10 (Ridge is a linear smoother). $\hat{\mathbf{Y}}_\lambda = \mathbf{S}_\lambda \mathbf{Y}$ with $\mathbf{S}_\lambda = \mathbf{X}(\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}'$, and

$$\text{df}(\lambda) = \text{tr}(\mathbf{S}_\lambda) = \sum_{j=1}^r \frac{d_j^2}{d_j^2 + \lambda}. \quad (10)$$

It decreases strictly from r at $\lambda = 0$ to 0 as $\lambda \rightarrow \infty$.

Proof. The form of $\mathbf{S}_\lambda$ is immediate from (2). By Theorem 6.6, $\mathbf{S}_\lambda = \sum_j \{d_j^2/(d_j^2 + \lambda)\} \mathbf{u}_j \mathbf{u}_j'$, and $\text{tr}(\mathbf{u}_j \mathbf{u}_j') = \mathbf{u}_j' \mathbf{u}_j = 1$ by the cyclic property of the trace. Each summand in $\text{df}(\lambda)$ is strictly decreasing in λ with limits 1 and 0. $\square$

Theorem 6.11 (Mallows for a linear smoother). Let $\mathbf{Y} = \boldsymbol{\mu} + \boldsymbol{\varepsilon}$ with $\mathbf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ and $\text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, let $\mathbf{Y}^{\text{new}} = \boldsymbol{\mu} + \boldsymbol{\varepsilon}^{\text{new}}$ be an independent replicate at the same design, and let $\hat{\mathbf{Y}} = \mathbf{S}\mathbf{Y}$. Then, with $\text{Err}_{\text{in}} = n^{-1} \mathbf{E} |\mathbf{Y}^{\text{new}} - \hat{\mathbf{Y}}|^2$ and $\text{SSE} = |\mathbf{Y} - \hat{\mathbf{Y}}|^2$,

$$n \text{Err}_{\text{in}} = \mathbf{E}[\text{SSE}] + 2\sigma^2 \text{tr}(\mathbf{S}). \quad (11)$$

Proof. Put $\mathbf{M} = \mathbf{I} - \mathbf{S}$. By the quadratic-form lemma of the prerequisites,

$$\mathbf{E}[\text{SSE}] = \mathbf{E} |\mathbf{M}\mathbf{Y}|^2 = \sigma^2 \text{tr}(\mathbf{M}'\mathbf{M}) + |\mathbf{M}\boldsymbol{\mu}|^2.$$

Since $\boldsymbol{\varepsilon}^{\text{new}}$ has mean zero and is independent of $\mathbf{Y}$,

$$n \text{Err}_{\text{in}} = n\sigma^2 + \mathbf{E} |\boldsymbol{\mu} - \mathbf{S}\mathbf{Y}|^2 = n\sigma^2 + \mathbf{E} |\mathbf{M}\boldsymbol{\mu} - \mathbf{S}\boldsymbol{\varepsilon}|^2 = n\sigma^2 + |\mathbf{M}\boldsymbol{\mu}|^2 + \sigma^2 \text{tr}(\mathbf{S}'\mathbf{S}),$$

using $\boldsymbol{\mu} - \mathbf{S}\mathbf{Y} = \mathbf{M}\boldsymbol{\mu} - \mathbf{S}\boldsymbol{\varepsilon}$ and $\mathbf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$. Subtracting and expanding $\text{tr}(\mathbf{M}'\mathbf{M}) = n - 2\text{tr}(\mathbf{S}) + \text{tr}(\mathbf{S}'\mathbf{S})$,

$$n \text{Err}_{\text{in}} - \mathbf{E}[\text{SSE}] = n\sigma^2 + \sigma^2 \text{tr}(\mathbf{S}'\mathbf{S}) - \sigma^2 (n - 2\text{tr}(\mathbf{S}) + \text{tr}(\mathbf{S}'\mathbf{S})) = 2\sigma^2 \text{tr}(\mathbf{S}). \quad \square$$

Remark. With $\mathbf{S} = \mathbf{H}$ the projection of Unit 2, $\text{tr}(\mathbf{H}) = d$ and (11) is the Mallows theorem of Unit 5. Nothing in the proof uses idempotency or symmetry, so the same accounting holds for ridge, and the price of a fit is $\text{tr}(\mathbf{S})$ rather than the number of coefficients estimated. Ridge estimates all k of them and pays $\text{df}(\lambda) \leq r$.

Proposition 6.12 (Self-influence formula). For a linear smoother, $\sum_{i=1}^n \text{cov}(\hat{Y}_i, Y_i) = \sigma^2 \text{tr}(\mathbf{S})$.

Proof. $\text{cov}(\hat{Y}_i, Y_i) = \text{cov}(\sum_l S_{il} Y_l, Y_i) = \sigma^2 S_{ii}$ because the Y_l are uncorrelated with common variance σ^2 . Summing over i gives the trace. $\square$

1.5 Why shrinkage can help

Proposition 6.13 (Bias and variance). Assume $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$, $\text{var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, and write $\mathbf{A}_\lambda = \mathbf{X}'\mathbf{X} + \lambda \mathbf{I}$. Then

$$\mathbf{E}[\hat{\boldsymbol{\beta}}_\lambda] - \boldsymbol{\beta} = -\lambda \mathbf{A}_\lambda^{-1} \boldsymbol{\beta}, \quad \text{var}(\hat{\boldsymbol{\beta}}_\lambda) = \sigma^2 \mathbf{A}_\lambda^{-1} \mathbf{X}'\mathbf{X} \mathbf{A}_\lambda^{-1}. \quad (12)$$

Proof. $\hat{\beta}_\lambda = \mathbf{A}_\lambda^{-1} \mathbf{X}' \mathbf{Y}$ is linear in $\mathbf{Y}$, so $\mathbb{E}[\hat{\beta}_\lambda] = \mathbf{A}_\lambda^{-1} \mathbf{X}' \mathbf{X} \beta = \mathbf{A}_\lambda^{-1} (\mathbf{A}_\lambda - \lambda \mathbf{I}) \beta = \beta - \lambda \mathbf{A}_\lambda^{-1} \beta$, and the variance is $\mathbf{A}_\lambda^{-1} \mathbf{X}' \text{var}(\mathbf{Y}) \mathbf{X} \mathbf{A}_\lambda^{-1}$ by the linear-map proposition of the prerequisites. $\square$

Theorem 6.14 (Some ridge always beats least squares). Assume in addition $\text{rank}(\mathbf{X}) = k$ and $\sigma^2 > 0$. Let $\gamma_1 \geq \dots \geq \gamma_k > 0$ be the eigenvalues of $\mathbf{X}' \mathbf{X}$ with orthonormal eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_k$, put $\alpha_j = \mathbf{v}_j' \beta$, and let $R(\lambda) = \mathbb{E} |\hat{\beta}_\lambda - \beta|^2$ be the total mean squared error. Then

$$R(\lambda) = \sum_{j=1}^k \frac{\sigma^2 \gamma_j}{(\gamma_j + \lambda)^2} + \sum_{j=1}^k \frac{\lambda^2 \alpha_j^2}{(\gamma_j + \lambda)^2}, \quad R'(\lambda) = \sum_{j=1}^k \frac{2\gamma_j(\lambda \alpha_j^2 - \sigma^2)}{(\gamma_j + \lambda)^3}. \quad (13)$$

Consequently R is strictly decreasing on $[0, \sigma^2 / \max_j \alpha_j^2)$, and in particular there is $\lambda > 0$ with $R(\lambda) < R(0)$: the least squares estimator is inadmissible under total squared error.

Proof. By the spectral theorem write $\mathbf{X}' \mathbf{X} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}'$ with $\mathbf{\Lambda} = \text{diag}(\gamma_j)$ and $\mathbf{V}$ orthogonal, so $\mathbf{A}_\lambda^{-1} = \mathbf{V}(\mathbf{\Lambda} + \lambda \mathbf{I})^{-1} \mathbf{V}'$. Applying the bias–variance decomposition of Unit 1 to each coordinate

and summing,

$$R(\lambda) = \sum_j \mathbb{E}(\hat{\beta}_{\lambda j} - \beta_j)^2 = \text{tr}\{\text{var}(\hat{\boldsymbol{\beta}}_\lambda)\} + |\mathbb{E}[\hat{\boldsymbol{\beta}}_\lambda] - \boldsymbol{\beta}|^2,$$

and by (12),

$$\text{tr}\{\text{var}(\hat{\boldsymbol{\beta}}_\lambda)\} = \sigma^2 \text{tr}\{(\mathbf{\Lambda} + \lambda \mathbf{I})^{-1} \mathbf{\Lambda} (\mathbf{\Lambda} + \lambda \mathbf{I})^{-1}\} = \sum_j \frac{\sigma^2 \gamma_j}{(\gamma_j + \lambda)^2},$$

using $\mathbf{V}'\mathbf{V} = \mathbf{I}$ and the cyclic property of the trace, while $|\lambda \mathbf{\Lambda}_\lambda^{-1} \boldsymbol{\beta}|^2 = \lambda^2 |(\mathbf{\Lambda} + \lambda \mathbf{I})^{-1} \mathbf{V}' \boldsymbol{\beta}|^2 = \sum_j \lambda^2 \alpha_j^2 / (\gamma_j + \lambda)^2$.

Differentiating term by term,

$$\frac{d}{d\lambda} \frac{\sigma^2 \gamma_j}{(\gamma_j + \lambda)^2} = \frac{-2\sigma^2 \gamma_j}{(\gamma_j + \lambda)^3}, \quad \frac{d}{d\lambda} \frac{\lambda^2 \alpha_j^2}{(\gamma_j + \lambda)^2} = \frac{2\lambda \alpha_j^2 (\gamma_j + \lambda) - 2\lambda^2 \alpha_j^2}{(\gamma_j + \lambda)^3} = \frac{2\lambda \alpha_j^2 \gamma_j}{(\gamma_j + \lambda)^3},$$

and adding gives the stated R' .

Every γ_j is positive, so each summand of $R'(\lambda)$ is negative as soon as $\lambda \alpha_j^2 < \sigma^2$ for every j , that is for $\lambda < \sigma^2 / \max_j \alpha_j^2$. At $\lambda = 0$ in particular $R'(0) = -2\sigma^2 \sum_j \gamma_j^{-2} < 0$. $\square$

Remark. The interval on which R decreases depends on σ^2 and on $\boldsymbol{\beta}$, both unknown, so the theorem does not deliver a usable λ ; it shows only that a beneficial one exists. The gain is largest when some γ_j is small, since the variance term $\sigma^2 \sum_j \gamma_j^{-1}$ at $\lambda = 0$ is then large while the bias introduced along that direction is bounded by α_j^2 . Choosing λ is left to cross-validation (§2.6).

2 The Lasso and the Elastic Net

ISLR §6.2.2–6.2.3.

2.1 Definition and basic structure

Definition 6.15 (Lasso). For $\lambda \geq 0$ the **lasso** criterion is

$$L_\lambda(\boldsymbol{\beta}) = |\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}|^2 + \lambda \|\boldsymbol{\beta}\|_1, \quad \|\boldsymbol{\beta}\|_1 = \sum_{j=1}^k |\beta_j|, \quad (14)$$

and a **lasso estimator** is a minimiser of L_λ over $\mathbb{R}^k$. The **active set** of a candidate $\boldsymbol{\beta}$ is $A(\boldsymbol{\beta}) = \{j : \beta_j \neq 0\}$.

Theorem 6.16 (Existence; the fit is unique even when the estimator is not). For $\lambda > 0$ the criterion L_λ attains its minimum. If $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\beta}}'$ are both minimisers then $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}\hat{\boldsymbol{\beta}}'$ and $\|\hat{\boldsymbol{\beta}}\|_1 = \|\hat{\boldsymbol{\beta}}'\|_1$.

Proof. L_λ is continuous, and $\|\boldsymbol{\beta}\|_1^2 = |\boldsymbol{\beta}|^2 + 2 \sum_{i < j} |\beta_i| |\beta_j| \geq |\boldsymbol{\beta}|^2$ gives $L_\lambda(\boldsymbol{\beta}) \geq \lambda |\boldsymbol{\beta}| \rightarrow \infty$, so L_λ is coercive and attains its minimum by the coercive-minimisation lemma of the prerequisites. Let $\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\beta}}'$ be minimisers with common value m , and suppose $\mathbf{X}\hat{\boldsymbol{\beta}} \neq \mathbf{X}\hat{\boldsymbol{\beta}}'$. Put $\boldsymbol{\beta}_t = t\hat{\boldsymbol{\beta}} + (1-t)\hat{\boldsymbol{\beta}}'$ for $t \in (0, 1)$. The map $\mathbf{w} \mapsto |\mathbf{Y} - \mathbf{w}|^2$ is strictly convex, so

$$|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}_t|^2 < t|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}|^2 + (1-t)|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}'|^2,$$

while $\|\boldsymbol{\beta}_t\|_1 \leq t\|\hat{\boldsymbol{\beta}}\|_1 + (1-t)\|\hat{\boldsymbol{\beta}}'\|_1$ by the triangle inequality. Adding, $L_\lambda(\boldsymbol{\beta}_t) < m$, which is impossible. Hence $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}\hat{\boldsymbol{\beta}}'$, the two sums of squares agree, and so do the two penalties. $\square$

Theorem 6.17 (The penalised and constrained forms). Let $P : \mathbb{R}^k \rightarrow [0, \infty)$ be convex with $P(\mathbf{0}) = 0$, and let $g(\boldsymbol{\beta}) = |\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}|^2$.

- (i) If $\hat{\boldsymbol{\beta}}$ minimises $g + \lambda P$ for some $\lambda \geq 0$, then $\hat{\boldsymbol{\beta}}$ minimises g subject to $P(\boldsymbol{\beta}) \leq P(\hat{\boldsymbol{\beta}})$.
- (ii) If $s > 0$ and $\hat{\boldsymbol{\beta}}$ minimises g subject to $P(\boldsymbol{\beta}) \leq s$, then there is $\lambda \geq 0$ such that $\hat{\boldsymbol{\beta}}$ minimises $g + \lambda P$ over $\mathbb{R}^k$.

Proof. (i) Let $P(\boldsymbol{\beta}) \leq P(\hat{\boldsymbol{\beta}})$. Then $g(\hat{\boldsymbol{\beta}}) + \lambda P(\hat{\boldsymbol{\beta}}) \leq g(\boldsymbol{\beta}) + \lambda P(\boldsymbol{\beta}) \leq g(\boldsymbol{\beta}) + \lambda P(\hat{\boldsymbol{\beta}})$, and cancelling gives $g(\hat{\boldsymbol{\beta}}) \leq g(\boldsymbol{\beta})$.

(ii) Apply the convex programme of the prerequisites with $\mathcal{D} = \mathbb{R}^k$, objective $f_0 = g$ and the single constraint $f_1(\boldsymbol{\beta}) = P(\boldsymbol{\beta}) - s$, both convex. The optimal value $p^* = g(\hat{\boldsymbol{\beta}})$ is finite, and Slater's condition holds at $\tilde{\boldsymbol{\beta}} = \mathbf{0}$, where $f_1(\mathbf{0}) = -s < 0$. By the strong duality theorem the dual optimum is attained at some $\lambda \geq 0$, and

$$g(\hat{\boldsymbol{\beta}}) = p^* = \inf_{\boldsymbol{\beta}} \{g(\boldsymbol{\beta}) + \lambda(P(\boldsymbol{\beta}) - s)\} \leq g(\hat{\boldsymbol{\beta}}) + \lambda(P(\hat{\boldsymbol{\beta}}) - s) \leq g(\hat{\boldsymbol{\beta}}),$$

the last step because $\lambda \geq 0$ and $P(\hat{\boldsymbol{\beta}}) \leq s$. Both inequalities are therefore equalities. Equality on the right gives $\lambda(P(\hat{\boldsymbol{\beta}}) - s) = 0$, and equality on the left gives $\inf_{\boldsymbol{\beta}} \{g(\boldsymbol{\beta}) + \lambda P(\boldsymbol{\beta})\} - \lambda s = g(\hat{\boldsymbol{\beta}})$.

Adding the first to the second,

$$\inf_{\boldsymbol{\beta}} \{g(\boldsymbol{\beta}) + \lambda P(\boldsymbol{\beta})\} = g(\hat{\boldsymbol{\beta}}) + \lambda P(\hat{\boldsymbol{\beta}}),$$

so the infimum is attained at $\hat{\boldsymbol{\beta}}$. □

Remark. Taking $P(\boldsymbol{\beta}) = \|\boldsymbol{\beta}\|_1$ gives the lasso in its two equivalent forms and $P(\boldsymbol{\beta}) = |\boldsymbol{\beta}|^2$ gives Proposition 6.3. The correspondence between λ and s is data-dependent: it runs through $s = P(\hat{\boldsymbol{\beta}}_\lambda)$ in one direction and through the multiplier of the constraint in the other.

2.2 Optimality conditions

Theorem 6.18 (Stationarity for the lasso). Let $\lambda > 0$ and $\mathbf{c} = \mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})$, so that $c_j = \mathbf{x}'_j(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})$. Then $\hat{\boldsymbol{\beta}}$ minimises L_λ if and only if

$$2c_j = \lambda \operatorname{sign}(\hat{\beta}_j) \quad \text{for } \hat{\beta}_j \neq 0, \quad |2c_j| \leq \lambda \quad \text{for } \hat{\beta}_j = 0. \quad (15)$$

Proof. L_λ is convex, so by the directional lemma of the prerequisites $\hat{\boldsymbol{\beta}}$ is a minimiser if and only if $L'_\lambda(\hat{\boldsymbol{\beta}}; \mathbf{v}) \geq 0$ for every $\mathbf{v} \in \mathbb{R}^k$. The first term of L_λ is differentiable with gradient $-2\mathbf{X}'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$, so it contributes $-2\mathbf{v}'\mathbf{c}$. For the second term, coordinate by coordinate,

$$\lim_{t \downarrow 0} \frac{|\hat{\beta}_j + tv_j| - |\hat{\beta}_j|}{t} = \begin{cases} v_j \operatorname{sign}(\hat{\beta}_j), & \hat{\beta}_j \neq 0, \\ |v_j|, & \hat{\beta}_j = 0, \end{cases}$$

the first case because the sign is unchanged for small t . Hence

$$L'_\lambda(\hat{\boldsymbol{\beta}}; \mathbf{v}) = \sum_{j: \hat{\beta}_j \neq 0} \{-2c_j + \lambda \operatorname{sign}(\hat{\beta}_j)\} v_j + \sum_{j: \hat{\beta}_j = 0} \{-2c_j v_j + \lambda |v_j|\}. \quad (16)$$

Suppose $\hat{\boldsymbol{\beta}}$ is a minimiser. Taking $\mathbf{v} = \pm \mathbf{e}_j$ with $\mathbf{e}_j$ the j -th standard basis vector: if $\hat{\beta}_j \neq 0$ the requirement is $\pm\{-2c_j + \lambda \operatorname{sign}(\hat{\beta}_j)\} \geq 0$, forcing $2c_j = \lambda \operatorname{sign}(\hat{\beta}_j)$; if $\hat{\beta}_j = 0$ it is $\mp 2c_j + \lambda \geq 0$, that is $|2c_j| \leq \lambda$.

Conversely, under (15) every term of the first sum in (16) vanishes and every term of the second is $\geq -|2c_j||v_j| + \lambda|v_j| \geq 0$, so $L'_\lambda(\hat{\boldsymbol{\beta}}; \mathbf{v}) \geq 0$ for all $\mathbf{v}$. $\square$

Corollary 6.19 (The smallest λ that kills every coefficient). $\mathbf{0}$ is a lasso estimator if and only if $\lambda \geq \lambda_{\max} := 2 \max_j |\mathbf{x}'_j \mathbf{Y}|$.

Proof. Put $\hat{\boldsymbol{\beta}} = \mathbf{0}$ in Theorem 6.18: every coefficient is zero, so the condition is $|2\mathbf{x}'_j \mathbf{Y}| \leq \lambda$ for every j . $\square$

Theorem 6.20 (Sparsity). For $\lambda > 0$ there is a lasso estimator $\hat{\boldsymbol{\beta}}$ whose active columns $\{\mathbf{x}_j : j \in A(\hat{\boldsymbol{\beta}})\}$ are linearly independent. In particular $|A(\hat{\boldsymbol{\beta}})| \leq \text{rank}(\mathbf{X}) \leq \min(n, k)$.

Proof. Among the minimisers choose one, $\hat{\boldsymbol{\beta}}$, and suppose its active columns are linearly dependent: there is $\mathbf{v} \neq \mathbf{0}$ with $v_j = 0$ for $j \notin A(\hat{\boldsymbol{\beta}})$ and $\mathbf{X}\mathbf{v} = \mathbf{0}$. For t in a neighbourhood of 0 every active coordinate $\hat{\beta}_j + tv_j$ keeps the sign of $\hat{\beta}_j$, so

$$\|\hat{\boldsymbol{\beta}} + t\mathbf{v}\|_1 = \sum_{j \in A} \text{sign}(\hat{\beta}_j)(\hat{\beta}_j + tv_j) = \|\hat{\boldsymbol{\beta}}\|_1 + tc, \quad c = \sum_{j \in A} \text{sign}(\hat{\beta}_j)v_j,$$

while $\mathbf{X}(\hat{\boldsymbol{\beta}} + t\mathbf{v}) = \mathbf{X}\hat{\boldsymbol{\beta}}$ leaves the sum of squares unchanged. Thus $L_\lambda(\hat{\boldsymbol{\beta}} + t\mathbf{v}) = L_\lambda(\hat{\boldsymbol{\beta}}) + \lambda tc$ near

$t = 0$, which is minimised at $t = 0$ only if $c = 0$; so $c = 0$ and L_λ is constant along the segment. Let $t^+ = \min\{-\hat{\beta}_j/v_j : j \in A, v_j \neq 0, -\hat{\beta}_j/v_j > 0\}$, finite because $\mathbf{v}$ has a nonzero coordinate in A and, replacing $\mathbf{v}$ by $-\mathbf{v}$ if necessary, at least one ratio is positive. On $[0, t^+]$ no active coordinate changes sign, so $L_\lambda(\hat{\beta} + t^+\mathbf{v}) = L_\lambda(\hat{\beta})$, and $\hat{\beta} + t^+\mathbf{v}$ is a minimiser whose active set is a proper subset of $A(\hat{\beta})$.

Repeating strictly decreases the active set, so the procedure stops, and it can only stop at a minimiser whose active columns are independent. A set of independent columns of $\mathbf{X}$ has at most $\text{rank}(\mathbf{X})$ members. $\square$

Remark. The bound is a genuine restriction only when $k > n$: a lasso fit can then involve at most n predictors however small λ is, whereas ridge always involves all k . It is a statement about the existence of a sparse solution, not about every solution; when the active columns are dependent the estimator is not unique and a solver may return a less sparse member of the same solution set, all of them sharing one fitted vector by Theorem 6.16.

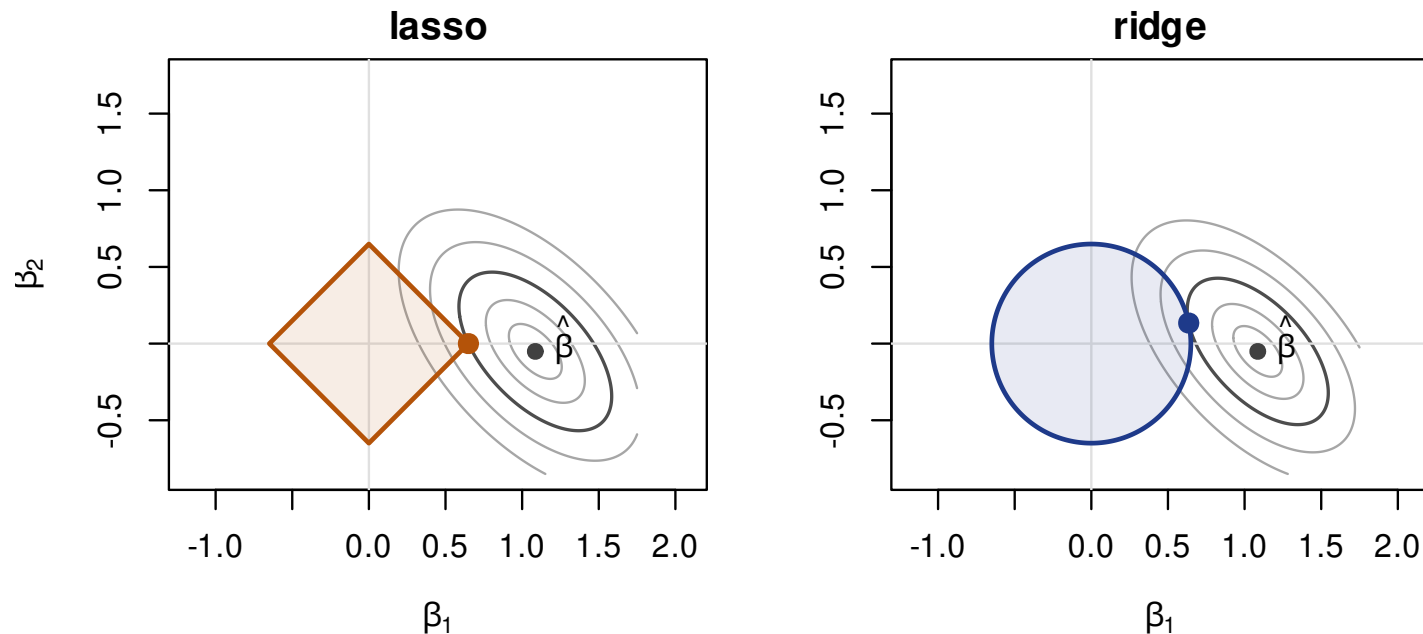


Figure 6.1: The constrained form of Theorem 6.17 at $k = 2$, with a common budget s . Grey curves are contours of $|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}|^2$ centred at $\hat{\boldsymbol{\beta}}$, the darker one passing through the solution. The ℓ_1 ball has corners on the axes, so the first contour to reach it touches a corner; the ℓ_2 ball is smooth.

2.3 The orthonormal design

Lemma 6.21 (Scalar soft thresholding). For $a \in \mathbb{R}$ and $\tau > 0$, the function $h(b) = (b-a)^2 + 2\tau|b|$ has the unique minimiser

$$S_\tau(a) = \text{sign}(a) (|a| - \tau)_+ = \begin{cases} a - \tau, & a > \tau, \\ 0, & |a| \leq \tau, \\ a + \tau, & a < -\tau. \end{cases} \quad (17)$$

Proof. h is continuous, coercive and strictly convex, so it has a unique minimiser by the coercive-minimisation lemma of the prerequisites. Since h with data $-a$ evaluated at $-b$ equals h with data a evaluated at b , it suffices to treat $a \geq 0$.

For $b < 0$ and $a \geq 0$, $(b-a)^2 = (|b|+a)^2 \geq (|b|-a)^2$, so $h(b) \geq h(|b|)$: the minimiser lies in $[0, \infty)$. There $h(b) = (b-a)^2 + 2\tau b$ is a parabola with vertex $a - \tau$, so its minimiser over $[0, \infty)$ is $\max(a - \tau, 0)$. $\square$

Theorem 6.22 (Ridge and lasso under an orthonormal design). Suppose $\mathbf{X}'\mathbf{X} = \mathbf{I}$, and write $\hat{\beta}_j = \mathbf{x}'_j \mathbf{Y}$ for the least squares coefficients. Then

$$\hat{\beta}_j^{\text{ridge}} = \frac{\hat{\beta}_j}{1 + \lambda}, \quad \hat{\beta}_j^{\text{lasso}} = S_{\lambda/2}(\hat{\beta}_j) = \text{sign}(\hat{\beta}_j) (|\hat{\beta}_j| - \tfrac{\lambda}{2})_+. \quad (18)$$

Proof. With $\mathbf{X}'\mathbf{X} = \mathbf{I}$ the least squares estimator is $\hat{\beta} = \mathbf{X}'\mathbf{Y}$ and, by Pythagoras applied to the orthogonal decomposition $\mathbf{Y} - \mathbf{X}\beta = (\mathbf{Y} - \mathbf{X}\hat{\beta}) + \mathbf{X}(\hat{\beta} - \beta)$,

$$|\mathbf{Y} - \mathbf{X}\beta|^2 = |\mathbf{Y} - \mathbf{X}\hat{\beta}|^2 + |\hat{\beta} - \beta|^2 = |\mathbf{Y} - \mathbf{X}\hat{\beta}|^2 + \sum_{j=1}^k (\beta_j - \hat{\beta}_j)^2.$$

Both criteria therefore separate into k scalar problems. The ridge problem $\min_b \{(b - \hat{\beta}_j)^2 + \lambda b^2\}$ has derivative $2(b - \hat{\beta}_j) + 2\lambda b$, vanishing at $b = \hat{\beta}_j / (1 + \lambda)$, and the objective is strictly convex. The lasso problem is Lemma 6.21 with $2\tau = \lambda$. $\square$

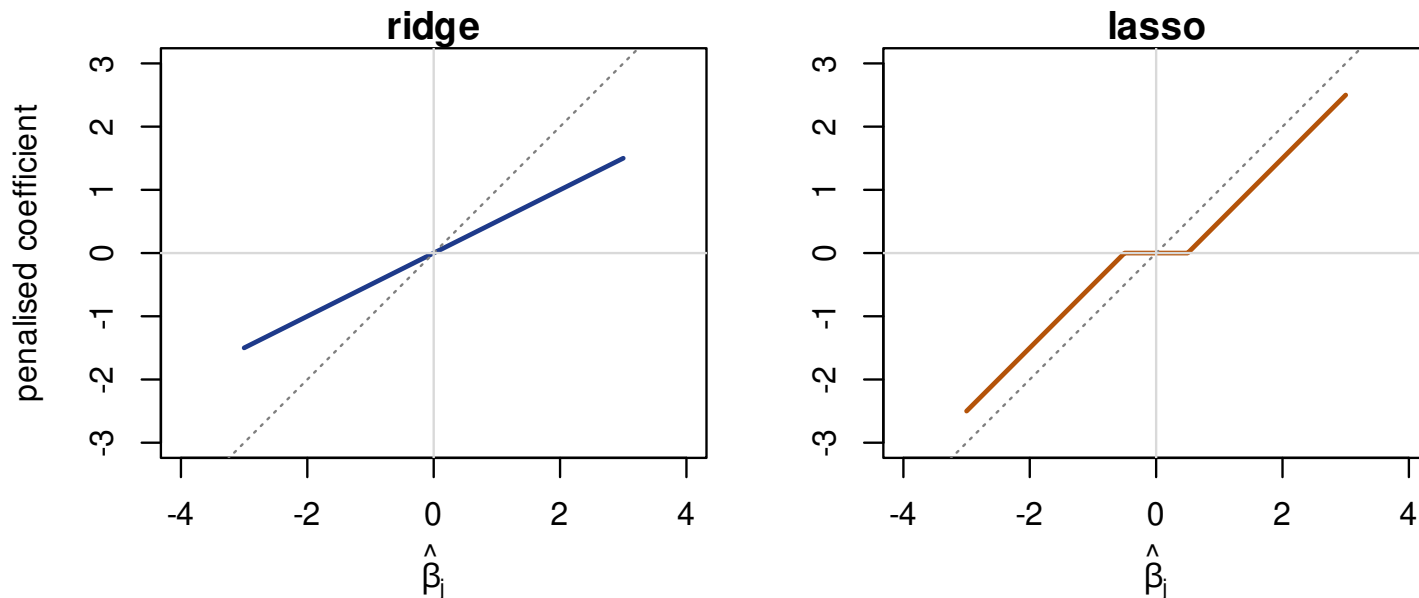


Figure 6.2: Shrinkage under an orthonormal design at $\lambda = 1$ (Theorem 6.22). The dotted line is the identity. Ridge multiplies every least squares coefficient by $1/(1 + \lambda)$; the lasso subtracts $\lambda/2$ from the modulus and sets to zero everything below that threshold.

Remark. Ridge shrinks by a constant factor, so no coefficient is ever exactly zero. The lasso shrinks by a constant amount and therefore zeroes every coefficient with $|\hat{\beta}_j| \leq \lambda/2$: selection and shrinkage happen in one operation. For a general design the two statements survive only qualitatively, and the exact description is Theorem 6.18.

2.4 Coordinate descent

Proposition 6.23 (The coordinate update). Fix j and all coordinates but the j -th, and let $\mathbf{r}^{(j)} = \mathbf{Y} - \sum_{l \neq j} \mathbf{x}_l \beta_l$ be the partial residual. Assume $\mathbf{x}_j \neq \mathbf{0}$. The minimiser of $\beta_j \mapsto L_\lambda(\boldsymbol{\beta})$ is

$$\beta_j \longleftarrow \frac{S_{\lambda/2}(\mathbf{x}_j' \mathbf{r}^{(j)})}{|\mathbf{x}_j|^2}. \quad (19)$$

Proof. As a function of β_j alone the criterion is, up to terms free of β_j ,

$$|\mathbf{r}^{(j)} - \mathbf{x}_j \beta_j|^2 + \lambda |\beta_j| = |\mathbf{x}_j|^2 \left(\beta_j - \frac{\mathbf{x}_j' \mathbf{r}^{(j)}}{|\mathbf{x}_j|^2} \right)^2 + \lambda |\beta_j| + \text{const.}$$

Dividing by $|\mathbf{x}_j|^2$ puts it in the form of Lemma 6.21 with $a = \mathbf{x}_j' \mathbf{r}^{(j)} / |\mathbf{x}_j|^2$ and $\tau = \lambda / (2|\mathbf{x}_j|^2)$, whose minimiser is $\text{sign}(a)(|a| - \tau)_+$. Multiplying numerator and denominator by $|\mathbf{x}_j|^2$ turns that into (19). $\square$

Theorem 6.24 (Coordinate-wise optimality is optimality). Suppose every column of $\mathbf{X}$ is nonzero and $\boldsymbol{\beta}$ satisfies (19) for every j simultaneously. Then $\boldsymbol{\beta}$ minimises L_λ .

Proof. Fix j and write $z = \mathbf{x}_j' \mathbf{r}^{(j)}$. Since $\mathbf{Y} - \mathbf{X}\boldsymbol{\beta} = \mathbf{r}^{(j)} - \mathbf{x}_j\beta_j$,

$$c_j = \mathbf{x}_j'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = z - |\mathbf{x}_j|^2\beta_j.$$

If $\beta_j = 0$ then (19) forces $|z| \leq \lambda/2$ and $c_j = z$, so $|2c_j| \leq \lambda$. If $\beta_j \neq 0$ then $|\mathbf{x}_j|^2\beta_j = \text{sign}(z)(|z| - \lambda/2)$ with $\text{sign}(\beta_j) = \text{sign}(z)$, whence

$$c_j = z - \text{sign}(z)|z| + \text{sign}(z)\frac{\lambda}{2} = \frac{\lambda}{2}\text{sign}(\beta_j).$$

So (15) holds at every coordinate, and Theorem 6.18 applies. $\square$

Remark. The two results together are an algorithm: sweep $j = 1, \dots, k$ applying the closed-form update (19), and stop when a sweep changes nothing. Each update is $O(n)$ once the residual is carried along, and a converged sweep certifies a global minimum by Theorem 6.24. This is what `glmnet` implements, over a decreasing grid of λ starting at $\lambda_{\max}$ of Corollary 6.19, where the solution is $\mathbf{0}$, and using the previous solution as the starting point at the next λ .

2.5 The elastic net

Definition 6.25 (Elastic net). For $\lambda_1, \lambda_2 \geq 0$ the **elastic net** criterion is

$$E(\boldsymbol{\beta}) = |\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}|^2 + \lambda_1 \|\boldsymbol{\beta}\|_1 + \lambda_2 |\boldsymbol{\beta}|^2. \quad (20)$$

Proposition 6.26 (Uniqueness). If $\lambda_2 > 0$ then E has exactly one minimiser.

Proof. $|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}|^2 + \lambda_1 \|\boldsymbol{\beta}\|_1$ is convex and $\lambda_2 |\boldsymbol{\beta}|^2$ is strictly convex, so E is strictly convex; it is coercive because $E(\boldsymbol{\beta}) \geq \lambda_2 |\boldsymbol{\beta}|^2$. Apply the coercive-minimisation lemma of the prerequisites. $\square$

Theorem 6.27 (Grouping). Suppose $\mathbf{x}_j = \mathbf{x}_l$ for some $j \neq l$.

- (i) If $\lambda_2 > 0$, the elastic net estimator satisfies $\hat{\beta}_j = \hat{\beta}_l$.
- (ii) If $\lambda_2 = 0$ and $\lambda_1 > 0$ and $\hat{\beta}$ is a lasso estimator with $\hat{\beta}_j \hat{\beta}_l \geq 0$, then replacing $(\hat{\beta}_j, \hat{\beta}_l)$ by $(t, \hat{\beta}_j + \hat{\beta}_l - t)$ gives another lasso estimator for every t between 0 and $\hat{\beta}_j + \hat{\beta}_l$.

Proof. Let π be the permutation of $\mathbb{R}^k$ that exchanges coordinates j and l . Since $\mathbf{x}_j = \mathbf{x}_l$, $\mathbf{X}\pi(\beta) = \mathbf{X}\beta$, and both penalties are invariant under π , so $E(\pi(\beta)) = E(\beta)$ for every β .

(i) By Proposition 6.26 the minimiser $\hat{\beta}$ is unique, and $\pi(\hat{\beta})$ is also a minimiser, so $\pi(\hat{\beta}) = \hat{\beta}$, that is $\hat{\beta}_j = \hat{\beta}_l$.

(ii) Write $\sigma = \hat{\beta}_j + \hat{\beta}_l$ and let β_t be $\hat{\beta}$ with the pair replaced by $(t, \sigma - t)$. Then $\mathbf{X}\beta_t = \mathbf{X}\hat{\beta}$, so the sums of squares agree. For t between 0 and σ both t and $\sigma - t$ have the sign of σ , so $|t| + |\sigma - t| = |\sigma|$; and $\hat{\beta}_j \hat{\beta}_l \geq 0$ gives $|\hat{\beta}_j| + |\hat{\beta}_l| = |\sigma|$ as well. The penalties therefore agree, and $E(\beta_t) = E(\hat{\beta})$.

The hypothesis in (ii) is automatic: $\mathbf{x}_j = \mathbf{x}_l$ makes $c_j = c_l$ in Theorem 6.18, and (15) then forces $\text{sign}(\hat{\beta}_j) = \text{sign}(\hat{\beta}_l)$ whenever both are nonzero. $\square$

Remark. Two perfectly correlated predictors are shared equally by the elastic net and split arbitrarily by the lasso: the whole segment in (ii) consists of minimisers, and which one a solver returns depends on the solver. Near-duplicates behave the same way to a good approximation, which is why a lasso active set is unstable under resampling when the predictors are strongly correlated (§5.3).

Proposition 6.28 (Elastic net under an orthonormal design). If $\mathbf{X}'\mathbf{X} = \mathbf{I}$ then $\hat{\beta}_j^{\text{enet}} = S_{\lambda_1/2}(\hat{\beta}_j)/(1 + \lambda_2)$.

Proof. As in Theorem 6.22 the problem separates, and the j -th scalar criterion is $(b - \hat{\beta}_j)^2 + \lambda_1|b| + \lambda_2 b^2$. Completing the square in the two quadratic terms,

$$(1 + \lambda_2) \left(b - \frac{\hat{\beta}_j}{1 + \lambda_2} \right)^2 + \lambda_1|b| + \text{const},$$

and dividing by $1 + \lambda_2$ gives Lemma 6.21 with $a = \hat{\beta}_j/(1 + \lambda_2)$ and $\tau = \lambda_1/\{2(1 + \lambda_2)\}$. Its minimiser is $\text{sign}(a)(|a| - \tau)_+ = S_{\lambda_1/2}(\hat{\beta}_j)/(1 + \lambda_2)$. $\square$

Remark. `glmnet` parametrises the same family by a mixing parameter $\alpha \in [0, 1]$ and one tuning constant, with $\alpha = 1$ the lasso and $\alpha = 0$ ridge. The translation between its constant and the λ of (14) and (1) is established numerically in §4.3.

2.6 Choosing the tuning parameter

The criteria of Unit 5 are unavailable for the lasso, since SSE is not a χ^2 multiple and the number of estimated parameters is not the right charge; and in the high-dimensional case of §3 the estimate $\hat{\sigma}^2$ they need does not exist. Cross-validation needs neither.

Definition 6.29 (Cross-validated tuning). Fix a grid $\lambda_1 > \dots > \lambda_G$ and a partition of $\{1, \dots, n\}$ into V folds $F_1, \dots, F_V$. Let $\hat{f}_\lambda^{(-v)}$ be the fit obtained from the data outside F_v at tuning constant λ . The **cross-validation curve** and its standard error are

$$\text{CV}(\lambda) = \frac{1}{n} \sum_{v=1}^V \sum_{i \in F_v} (Y_i - \hat{f}_\lambda^{(-v)}(x_i))^2, \quad \text{SE}(\lambda) = \sqrt{\frac{1}{V-1} \sum_{v=1}^V \frac{|F_v|}{n} (e_v(\lambda) - \text{CV}(\lambda))^2}, \quad (21)$$

where $e_v(\lambda) = |F_v|^{-1} \sum_{i \in F_v} (Y_i - \hat{f}_\lambda^{(-v)}(x_i))^2$ is the v -th fold's mean error. Two selections are in

use:

$$\lambda_{\min} = \arg \min_{\lambda} \text{CV}(\lambda), \quad \lambda_{1\text{SE}} = \max\{\lambda : \text{CV}(\lambda) \leq \text{CV}(\lambda_{\min}) + \text{SE}(\lambda_{\min})\}. \quad (22)$$

Remark. CV is an average of n terms and SE an estimate of its standard deviation from V fold means, so $\lambda_{1\text{SE}}$ is the most heavily penalised model whose error is within sampling noise of the best. It is the more parsimonious of the two, and by Theorem 6.18 a larger λ can only turn coefficients off, so it selects a subset of what $\lambda_{\min}$ selects whenever the active sets are nested along the path.

Remark. The grid runs from $\lambda_{\max}$ of Corollary 6.19, at which the fit is $\mathbf{0}$, down to a small multiple of it, equally spaced on the log scale. Everything used in the fit, standardisation included, must be recomputed inside each fold: the pitfall of Unit 1 about selecting predictors before cross-validating applies unchanged here.

3 High Dimensions and Nearest Neighbours

ISLR §6.4, §3.5.

3.1 Least squares when $k \geq n$

Let $\mathbf{X}_+ = [\mathbf{1} \ \mathbf{X}]$ be the design with its intercept column, of size $n \times (k+1)$, and let $\mathbf{H}$ be the projection onto $\mathcal{C}(\mathbf{X}_+)$.

Theorem 6.30 (A perfect fit carries no information). If $\text{rank}(\mathbf{X}_+) = n$ then $\mathbf{H} = \mathbf{I}$, so every residual is zero, $\text{SSE} = 0$ and $R^2 = 1$, whatever $\mathbf{Y}$ is.

Proof. $\mathcal{C}(\mathbf{X}_+)$ is a subspace of $\mathbb{R}^n$ of dimension n , hence all of $\mathbb{R}^n$. The orthogonal projection onto $\mathbb{R}^n$ is the identity, so $\hat{\mathbf{Y}} = \mathbf{Y}$ and $\text{SSE} = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 = 0$; and $R^2 = 1 - \text{SSE}/\text{SST} = 1$ whenever $\text{SST} > 0$. $\square$

Corollary 6.31. Under the hypothesis of Theorem 6.30 with $\text{rank}(\mathbf{X}_+) = n = k+1$: the residual variance estimate $\hat{\sigma}^2 = \text{SSE}/(n-k-1)$ is $0/0$, and C_p , AIC, BIC and $\bar{R}^2$, all of which are built from SSE and $\hat{\sigma}^2$, are undefined. With $d < n$ parameters they are defined, but R^2 is non-decreasing in d and reaches 1 at $d = n$ whatever the response, so it ranks every nested sequence in favour of the largest model.

Proof. The first statement is Theorem 6.30 together with $n - k - 1 = 0$. The second is the monotonicity of R^2 from Unit 2, applied along a nested sequence of designs whose column space grows to $\mathbb{R}^n$, where Theorem 6.30 gives $\text{SSE} = 0$. $\square$

Proposition 6.32 (Non-uniqueness). Let $\hat{\boldsymbol{\beta}}$ satisfy the normal equations $\mathbf{X}'_+ \mathbf{X}_+ \hat{\boldsymbol{\beta}} = \mathbf{X}'_+ \mathbf{Y}$. The set of least squares solutions is $\hat{\boldsymbol{\beta}} + \mathcal{N}(\mathbf{X}_+)$, an affine subspace of dimension $k + 1 - \text{rank}(\mathbf{X}_+)$, and every member gives the same fitted vector. If $k + 1 > n$ this dimension is at least $k + 1 - n > 0$.

Proof. $|\mathbf{Y} - \mathbf{X}_+ \boldsymbol{\beta}|^2$ depends on $\boldsymbol{\beta}$ only through $\mathbf{X}_+ \boldsymbol{\beta}$, and the minimum is attained exactly when $\mathbf{X}_+ \boldsymbol{\beta} = \mathbf{H}\mathbf{Y}$ by the projection theorem of Unit 2. The solution set of $\mathbf{X}_+ \boldsymbol{\beta} = \mathbf{H}\mathbf{Y}$ is $\hat{\boldsymbol{\beta}} + \mathcal{N}(\mathbf{X}_+)$, whose dimension is $k + 1 - \text{rank}(\mathbf{X}_+)$ by rank-nullity, and $\text{rank}(\mathbf{X}_+) \leq n$ always. $\square$

Remark. R signals the degeneracy by returning NA for $k + 1 - \text{rank}(\mathbf{X}_+)$ of the coefficients: it fits a maximal independent subset of the columns and refuses to report the rest. The reported model is one arbitrary member of the affine family of Proposition 6.32.

3.2 Regularised fitting in high dimensions

Ridge and the lasso are defined without any rank condition, and the results already proved say precisely how they behave when k is large.

Proposition 6.33. For $\lambda > 0$ and any n, k :

- (i) the ridge estimator exists and is unique (Theorem 6.2), and its effective degrees of freedom satisfy $\text{df}(\lambda) < \text{rank}(\mathbf{X}) \leq \min(n, k)$ (Proposition 6.10);
- (ii) a lasso estimator exists, all of them share one fitted vector (Theorem 6.16), and one of them has at most $\text{rank}(\mathbf{X}) \leq \min(n, k)$ nonzero coefficients (Theorem 6.20);
- (iii) if $k > n$ the columns of $\mathbf{X}$ are linearly dependent, so some predictor is an exact linear combination of the others.

Proof. (i) and (ii) restate the cited results, using $\text{rank}(\mathbf{X}) \leq \min(n, k)$ and, in (i), $d_j^2/(d_j^2 + \lambda) < 1$ for every j when $\lambda > 0$. For (iii), k vectors in $\mathbb{R}^n$ with $k > n$ are dependent, so a nontrivial combination vanishes and any column with a nonzero coefficient in it is a combination of the others. $\square$

Remark. Part (iii) is the exact form of the warning that a selected set of predictors is not identifiable when $k > n$: a different sample, or a different solver, can return a disjoint set with the same fitted values. Reported error must then come from data untouched by the fitting, and SSE, R^2 and the p -values of Unit 3 computed on the training data are worthless by Theorem 6.30.

3.3 Nearest neighbour regression

Definition 6.34 (KNN regression). Given training pairs $(x_1, Y_1), \dots, (x_n, Y_n)$ with $x_i \in \mathbb{R}^k$, an integer $K \leq n$ and a point x_0 , let $\mathcal{N}_0 = \mathcal{N}_K(x_0)$ be the set of indices of the K training inputs nearest x_0 in Euclidean distance. The **KNN regression estimate** is

$$\hat{f}(x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} Y_i. \quad (23)$$

Theorem 6.35 (Pointwise error). Treat the inputs as fixed and let $Y_i = f(x_i) + \varepsilon_i$ with the ε_i uncorrelated, mean zero and variance σ^2 . Let $Y_0 = f(x_0) + \varepsilon_0$ be a new observation at x_0 ,

uncorrelated with the training errors. Then

$$\mathbb{E}[(Y_0 - \hat{f}(x_0))^2] = \underbrace{\sigma^2}_{\text{irreducible}} + \underbrace{\left(f(x_0) - \frac{1}{K} \sum_{i \in \mathcal{N}_0} f(x_i)\right)^2}_{\text{bias}^2} + \underbrace{\frac{\sigma^2}{K}}_{\text{variance}}. \quad (24)$$

Proof. The inputs being fixed, $\mathcal{N}_0$ is a fixed index set, and $\hat{f}(x_0)$ is an average of K uncorrelated variables, so

$$\mathbb{E}[\hat{f}(x_0)] = \frac{1}{K} \sum_{i \in \mathcal{N}_0} f(x_i), \quad \text{var}(\hat{f}(x_0)) = \frac{\sigma^2}{K}.$$

Since ε_0 is uncorrelated with $\hat{f}(x_0)$ and has mean zero,

$$\mathbb{E}[(Y_0 - \hat{f}(x_0))^2] = \sigma^2 + \mathbb{E}[(f(x_0) - \hat{f}(x_0))^2] = \sigma^2 + (f(x_0) - \mathbb{E}[\hat{f}(x_0)])^2 + \text{var}(\hat{f}(x_0)),$$

the last step being the bias–variance decomposition of Unit 1. □

Proposition 6.36 (Average variance of the two fits). Let $\mathbf{X}_+$ have full column rank $k + 1$ and let $\hat{Y}_i$ be the least squares fitted values. Then

$$\frac{1}{n} \sum_{i=1}^n \text{var}(\hat{Y}_i) = \frac{\sigma^2(k+1)}{n}, \quad \text{while} \quad \frac{1}{n} \sum_{i=1}^n \text{var}(\hat{f}(x_i)) = \frac{\sigma^2}{K} \quad (25)$$

for the KNN fit at the training inputs. The two are equal at $K = n/(k+1)$.

Proof. $\text{var}(\hat{Y}_i) = \sigma^2 h_{ii}$ by Unit 2, and $\sum_i h_{ii} = \text{tr}(\mathbf{H}) = k+1$ because $\mathbf{H}$ is a projection of rank $k+1$. The KNN statement is the variance computed in the proof of Theorem 6.35, which does not depend on i . $\square$

Remark. The variance of the nearest neighbour fit does not involve k at all. Everything the dimension does to KNN it does through the bias term of (24), which is controlled by how far the K nearest inputs actually are from x_0 .

3.4 The curse of dimensionality

Theorem 6.37 (Distance to the nearest neighbours). Let $x_1, \dots, x_n$ be independent and uniform on the cube $C = [-\frac{1}{2}, \frac{1}{2}]^k$, and let R_K be the K -th smallest of $\|x_1\|_\infty, \dots, \|x_n\|_\infty$, the sup-norm distance from the centre of C to its K -th nearest neighbour among the sample. For $0 \leq r \leq \frac{1}{2}$,

$$\mathbf{P}\{R_K \leq r\} = \mathbf{P}\{\text{Bin}(n, (2r)^k) \geq K\}. \quad (26)$$

In particular the median of R_1 is

$$m(n, k) = \frac{1}{2} (1 - 2^{-1/n})^{1/k}. \quad (27)$$

Proof. The coordinates $x_{i1}, \dots, x_{ik}$ are independent and uniform on $[-\frac{1}{2}, \frac{1}{2}]$, so

$$\mathbf{P}\{\|x_i\|_\infty \leq r\} = \prod_{j=1}^k \mathbf{P}\{|x_{ij}| \leq r\} = (2r)^k.$$

The events are independent across i , so the number of points within distance r is binomial with n

trials and success probability $(2r)^k$, and $R_K \leq r$ exactly when that count is at least K .

For $K = 1$ this reads $\mathbf{P}\{R_1 > r\} = \{1 - (2r)^k\}^n$. Setting the right side to $\frac{1}{2}$ gives $(2r)^k = 1 - 2^{-1/n}$ and hence (27). $\square$

Proposition 6.38 (Neighbourhoods are not local). A cube centred at the origin, with side ℓ , contained in C , carries probability ℓ^k . To carry a fraction p of the distribution it must have side $\ell = p^{1/k}$.

Proof. Independence of the coordinates again: the probability is $\prod_{j \leq k} \ell = \ell^k$. $\square$

Corollary 6.39 (Neighbourhood radius bounds the bias). Suppose $|f(u) - f(v)| \leq L\|u - v\|_\infty$ for all $u, v \in \mathbb{R}^k$. Then the bias term of (24) satisfies

$$\left| f(x_0) - \frac{1}{K} \sum_{i \in \mathcal{N}_0} f(x_i) \right| \leq L \max_{i \in \mathcal{N}_0} \|x_i - x_0\|_\infty. \quad (28)$$

Proof. The left side is $|K^{-1} \sum_{i \in \mathcal{N}_0} \{f(x_0) - f(x_i)\}|$, at most the average of $L\|x_0 - x_i\|_\infty$ over $i \in \mathcal{N}_0$, itself at most the maximum. $\square$

Remark. Equation (27) is the quantitative content of the phrase **curse of dimensionality**: with $n = 500$ points in $k = 10$ dimensions the typical nearest neighbour of the centre sits at sup-norm distance 0.259, over half the way to the boundary of a cube of half-side 0.5, whereas in one dimension it sits at 0.0007. The neighbours that enter (23) are then not near, the bound (28) is weak, and the average in (23) estimates something other than $f(x_0)$. Increasing n is almost no help: $m(n, k)$ falls only like $n^{-1/k}$, so at $k = 10$ raising n from 100 to 5000 moves the typical distance only from 0.304 to 0.206.

3.5 Nearest neighbours against least squares

Remark. Collecting the three results: least squares carries a variance $\sigma^2(k+1)/n$ and a bias that is zero exactly when f is linear in the given predictors; KNN carries a variance σ^2/K that ignores k and a bias governed by the neighbourhood radius, which by Theorem 6.37 grows quickly with k . So least squares wins whenever f is close to linear, and KNN wins when f is far from linear and

k is small. Adding predictors that do not enter f costs least squares σ^2/n per predictor and costs KNN a deterioration in the bias with no compensating gain, which is why the crossover in §4.6 occurs at a small k . When the two are close, least squares is preferred for the inferential machinery of Units 2 and 3, which KNN does not have.

4 Computation: Formulas Against Package Output

ISLR §6.5, §3.6.

Data: **Hitters** from ISLR2, salary on the other variables, rows with missing salary removed. The design is centred and scaled by (5).

```
1 suppressMessages({library(ISLR2); library(MASS); library(glmnet); library(FNN)})
2 d <- na.omit(Hitters); n <- nrow(d)
3 X <- model.matrix(Salary ~ ., d)[, -1]
4 sdn <- function(v) sqrt(mean((v - mean(v))^2)) # divisor n
5 Xs <- scale(X, center = TRUE, scale = apply(X, 2, sdn))
6 y <- d$Salary; yc <- y - mean(y); k <- ncol(Xs)
7 c(n = n, k = k, max_col_mean = max(abs(colMeans(Xs))),
8   max_col_ss_less_n = max(abs(colSums(Xs^2) - n)))
```

```
##              n              k      max_col_mean
##      2.630e+02      1.900e+01      1.220e-16
## max_col_ss_less_n
##      5.684e-14
```

4.1 The ridge estimator (§1.1, §1.3)

Three routes to the same vector: the linear solve of (2), the singular value expansion of Theorem 6.6, and `MASS::lm.ridge`, whose coefficients are reported on the original scale and are put back on the standardised one.

```
1 lam      <- 100
2 b_solve <- drop(solve(crossprod(Xs) + lam * diag(k), crossprod(Xs, yc)))
3 sv      <- svd(Xs); dj <- sv$d
4 b_svd   <- drop(sv$v %*% ((dj / (dj^2 + lam)) * crossprod(sv$u, yc)))
5 b_pkg   <- coef(lm.ridge(Salary ~ ., data = d, lambda = lam))[-1] *
6         apply(X, 2, sdn)
7 c(solve_vs_svd = max(abs(b_solve - b_svd)),
8   solve_vs_lm.ridge = max(abs(b_solve - b_pkg)),
9   coefficient_scale = max(abs(b_solve)))
```

```
##      solve_vs_svd solve_vs_lm.ridge coefficient_scale
```

##	1.136e-13	1.349e-13	5.667e+01
----	-----------	-----------	-----------

Corollary 6.7 says the ridge fit is the least squares fit on the principal component scores with the j -th coefficient multiplied by $d_j^2/(d_j^2 + \lambda)$.

```

1 Z <- Xs %*% sv$v                                # score vectors, Unit 5
2 th <- drop(crossprod(Z, yc) / colSums(Z^2))      # their OLS coefficients
3 c(pc_form_vs_ridge_fit =
4   max(abs(drop(Z %*% (dj^2/(dj^2 + lam) * th)) - drop(Xs %*% b_solve))),
5   pc_form_vs_ols_fit =
6   max(abs(drop(Z %*% th) - fitted(lm(yc ~ Xs - 1)))),
7   shrinkage_largest_d = (dj^2/(dj^2 + lam))[1],
8   shrinkage_smallest_d = (dj^2/(dj^2 + lam))[k])

```

##	pc_form_vs_ridge_fit	pc_form_vs_ols_fit
##	6.253e-13	7.162e-12
##	shrinkage_largest_d	shrinkage_smallest_d
##	9.504e-01	3.113e-03

The direction of least variation is shrunk by a factor of 0.003 and the direction of most variation by a factor of 0.95. Proposition 6.8 is the same fact read on the spectrum, and Unit 4's variance inflation

factor is the same near-degeneracy read one predictor at a time.

```
1 c(kappa_XtX = max(dj^2)/min(dj^2),
2   kappa_ridge = (max(dj^2) + lam)/(min(dj^2) + lam),
3   largest_VIF = max(car::vif(lm(Salary ~ ., data = d))))
```

```
##      kappa_XtX kappa_ridge largest_VIF
##      6131.34      20.08      502.95
```

4.2 Effective degrees of freedom (§1.4)

$\text{tr}(\mathbf{S}_\lambda)$ computed from the matrix, from the singular values by (10), and from the covariance identity of Proposition 6.12 estimated by simulation.

```
1 S <- Xs %*% solve(crossprod(Xs) + lam * diag(k)) %*% t(Xs)
2 set.seed(1); B <- 4000; sg <- 200; mu <- drop(Xs %*% b_solve)
3 Ys <- replicate(B, mu + rnorm(n, sd = sg)); Fs <- S %*% Ys
4 cv <- sum(rowMeans((Fs - rowMeans(Fs)) * (Ys - rowMeans(Ys)))) * B/(B - 1)
5 c(trace_S = sum(diag(S)), sum_of_shrinkages = sum(dj^2/(dj^2 + lam)),
6   simulated_cov_identity = cv/sg^2, df_at_lambda_zero = sum(dj^2/dj^2),
7   k = k)
```

```
##      trace_S      sum_of_shrinkages
```

```
##                7.960                7.960
## simulated_cov_identity      df_at_lambda_zero
##                7.921                19.000
##                k
##                19.000
```

Theorem 6.11 states that the training error understates in-sample prediction error by exactly $2\sigma^2 \text{tr}(\mathbf{S})$, whether or not $\mathbf{S}$ is a projection.

```
1 set.seed(1); B <- 20000; mu <- drop(Xs %*% rep(20, k))
2 r <- replicate(B, {
3   yy <- mu + rnorm(n, sd = sg); fit <- drop(S %*% yy)
4   c(sse = sum((yy - fit)^2),
5     pred = sum((mu + rnorm(n, sd = sg) - fit)^2)) })
6 dif <- r["pred", ] - r["sse", ]
7 c(E_SSE = mean(r["sse", ]), n_Err_in = mean(r["pred", ]),
8   difference = mean(dif), mc_se = sd(dif)/sqrt(B),
9   two_sigma2_trace_S = 2 * sg^2 * sum(diag(S)))
```

```
##                E_SSE                n_Err_in                difference
##                10162493                10798668                636175
##                mc_se two_sigma2_trace_S
```

```
##                9269                636835
```

4.3 Mean squared error against λ (§1.5)

Fixing a β and a σ^2 makes $R(\lambda)$ of Theorem 6.14 computable in closed form.

```
1 gam <- dj^2; A <- crossprod(Xs)
2 set.seed(3); bb <- rnorm(k, sd = 20); s2 <- 200^2
3 al <- drop(crossprod(sv$v, bb))
4 R <- function(l) sum(s2*gam/(gam + l)^2) + sum(l^2*al^2/(gam + l)^2)
5 rbind(lambda = c(0, 10, 50, 100, 250, 500),
6       MSE     = round(sapply(c(0, 10, 50, 100, 250, 500), R)))
```

```
##           [,1] [,2] [,3] [,4] [,5] [,6]
## lambda      0   10   50  100  250  500
## MSE    187425 10096 4391 3572 3280 3407
```

```
1 opt <- optimize(R, c(0, 3000))
2 c(R_prime_at_0 = sum(2*gam*(0*al^2 - s2)/gam^3),
3   minus_2_sigma2_sum_gamma_sq = -2*s2*sum(gam^-2),
4   argmin = opt$minimum, minimum = opt$objective,
5   guaranteed_decrease_up_to = s2/max(al^2))
```

```
##          R_prime_at_0 minus_2_sigma2_sum_gamma_sq
##          -877745.46          -877745.46
##          argmin          minimum
##          244.21          3279.42
##  guaranteed_decrease_up_to
##          25.11
```

Least squares sits at the left end of the table; the best ridge is a factor of 57 below it. Simulation confirms both ends.

```
1  set.seed(4); Bs <- 4000; l0 <- 50
2  W <- solve(A + 10*diag(k)) %*% t(Xs); mu2 <- drop(Xs %*% bb)
3  sim <- replicate(Bs, {
4    yy <- mu2 + rnorm(n, sd = sqrt(s2))
5    c(ridge = sum((drop(W %*% yy) - bb)^2),
6      ols   = sum((drop(solve(A, crossprod(Xs, yy))) - bb)^2)) })
7  c(sim_ridge = mean(sim["ridge", ]), formula_ridge = R(10),
8    mc_se_ridge = sd(sim["ridge", ])/sqrt(Bs),
9    sim_ols = mean(sim["ols", ]), formula_ols = R(0),
10   mc_se_ols = sd(sim["ols", ])/sqrt(Bs))
```

```
##          sim_ridge formula_ridge  mc_se_ridge          sim_ols
```

```
##          4395.35          4391.23          15.62          183644.31
## formula_ols      mc_se_ols
##          187424.77          2814.24
```

4.4 The lasso (§2)

Under an orthonormal design Theorem 6.22 gives both estimators in closed form; a general-purpose minimiser reaches the same value.

```
1 soft <- function(a, tau) sign(a) * pmax(abs(a) - tau, 0)
2 Q  <- qr.Q(qr(Xs)); bo <- drop(crossprod(Q, yc)); lm4 <- 400
3 oL <- function(b) sum((yc - Q %*% b)^2) + lm4 * sum(abs(b))
4 oR <- function(b) sum((yc - Q %*% b)^2) + lm4 * sum(b^2)
5 ct <- list(maxit = 5000, reltol = 1e-14)
6 c(lasso_formula = oL(soft(bo, lm4/2)),
7   lasso_numeric = optim(bo, oL, method = "BFGS", control = ct)$value,
8   ridge_formula = oR(bo/(1 + lm4)),
9   ridge_numeric = optim(bo, oR, method = "BFGS", control = ct)$value,
10  zeroed_by_lasso = sum(soft(bo, lm4/2) == 0), of = k)
```

```
## lasso_formula lasso_numeric ridge_formula
##          30900558          30900558          53246498
```

##	ridge_numeric	zeroed_by_lasso	of
##	53246498	3	19

For a general design, coordinate descent (Proposition 6.23) run to convergence, then checked against the stationarity conditions (15).

```
1 lasso_cd <- function(X, y, lam, tol = 1e-12) {  
2   b <- numeric(ncol(X)); ss <- colSums(X^2); r <- y  
3   repeat {  
4     old <- b  
5     for (j in seq_along(b)) {  
6       r <- r + X[, j] * b[j]  
7       b[j] <- soft(sum(X[, j] * r), lam/2) / ss[j]  
8       r <- r - X[, j] * b[j]  
9     }  
10    if (max(abs(b - old)) < tol) break  
11  }  
12  b  
13 }  
14 lam1 <- 1000; bcd <- lasso_cd(Xs, yc, lam1)  
15 cj <- drop(crossprod(Xs, yc - Xs %*% bcd)); on <- bcd != 0
```

```

16 c(active = sum(on), of = k,
17   active_kkt_residual = max(abs(2*cj[on] - lam1*sign(bcd[on]))),
18   inactive_max_2c = max(abs(2*cj[!on])), lambda = lam1,
19   lambda_max = 2*max(abs(crossprod(Xs, yc))))

```

```

##           active           of active_kkt_residual
##      1.500e+01      1.900e+01      4.212e-10
## inactive_max_2c      lambda      lambda_max
##      9.232e+02      1.000e+03      1.343e+05

```

The active coordinates satisfy the equality in (15) to 4×10^{-10} and the inactive ones satisfy the inequality, the largest reaching 923 against $\lambda = 1000$. The threshold $\lambda_{\max}$ of Corollary 6.19 is two orders of magnitude above the λ used, so the fit is far from the null model.

glmnet minimises $(2n)^{-1}|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}|^2$ plus its own penalty, and for the gaussian family it also rescales the response internally. The translation is therefore not the same for the two ends of the family, and is established here rather than assumed: the lasso's λ of (14) is $2n$ times the reported one, while the ridge λ of (1) is n times the reported one divided by the standard deviation of $\mathbf{Y}$ computed with divisor n .

```

1 gl <- glmnet(Xs, yc, alpha = 1, lambda = lam1/(2*n) * c(4, 2, 1),

```

```

2      standardize = FALSE, intercept = FALSE, thresh = 1e-16)
3 bg <- as.matrix(coef(gl))[-1, 3]
4 lg <- 0.38
5 rg <- glmnet(Xs, yc, alpha = 0, lambda = lg * c(4, 2, 1),
6             standardize = FALSE, intercept = FALSE, thresh = 1e-16)
7 br <- as.matrix(coef(rg))[-1, 3]
8 imp <- drop(crossprod(Xs, yc - Xs %*% br)) / br      # ridge normal equations
9 c(cd_vs_glmnet = max(abs(bcd - bg)), coefficient_scale = max(abs(bcd)),
10   lasso_lambda = lam1, two_n_times_reported = 2*n*(lam1/(2*n)))

```

```

##      cd_vs_glmnet      coefficient_scale
##      0.0002788        276.7203386
##      lasso_lambda two_n_times_reported
##      1000.0000000        1000.0000000

```

```

1 c(ridge_implied_lambda = mean(imp), spread_over_coordinates = sd(imp),
2   n_times_reported = n*lg, over_sd_of_y = n*lg/sdn(y), sd_n_of_y = sdn(y))

```

```

##      ridge_implied_lambda spread_over_coordinates
##      0.221958623          0.000008877
##      n_times_reported      over_sd_of_y

```

```
##          99.940000000          0.221960535
##          sd_n_of_y
##          450.260223824
```

Dividing the ridge normal equations coordinate by coordinate must give the same number at every j , and it does, to within the solver's tolerance: that is Theorem 6.2 read backwards from the package's answer, and it identifies the penalty the package actually applied without relying on any documented conversion.

Duplicating a column separates the elastic net from the lasso exactly as Theorem 6.27 says.

```
1 Xd <- cbind(Xs, Hits2 = Xs[, "Hits"])
2 t(sapply(c(0, 0.5, 1), function(al) {
3   f <- glmnet(Xd, yc, alpha = al, lambda = 2*c(4, 2, 1), standardize = FALSE,
4             intercept = FALSE, thresh = 1e-16)
5   b <- unname(as.matrix(coef(f))[-1, 3])
6   c(alpha = al, Hits = b[2], Hits_copy = b[20],
7     difference = b[2] - b[20], total = b[2] + b[20]) })))
```

```
##      alpha Hits Hits_copy difference total
## [1,]  0.0 152.4   152.438   0.000185 304.9
## [2,]  0.5 145.4   145.449  -0.001899 290.9
```

```
| ## [3,]    1.0 270.6      4.685 265.871844 275.2
```

4.5 Cross-validation for λ (§2.6)

`cv.glmnet` refits each fold on **that fold's own** grid of tuning constants and then interpolates onto the full-data grid; reproducing it means doing the same.

```
1 set.seed(1); fold <- sample(rep(1:10, length.out = n))
2 cvg <- cv.glmnet(X, y, alpha = 1, foldid = fold); lgrid <- cvg$lambda
3 P <- matrix(NA_real_, n, length(lgrid))
4 for (v in 1:10) {
5   f <- glmnet(X[fold != v, ], y[fold != v], alpha = 1)      # its own grid
6   P[fold == v, ] <- predict(f, X[fold == v, ], drop = FALSE, s = lgrid)
7 }
8 cvm <- colMeans((y - P)^2)
9 fm  <- t(sapply(1:10, function(v)
10      colMeans((y[fold == v] - P[fold == v, , drop = FALSE])^2)))
11 cvsd <- sqrt(apply(scale(fm, cvm, FALSE)^2, 2, weighted.mean,
12      w = as.vector(table(fold)))/9)
13 i <- which.min(cvm)
14 c(max_diff_cv_curve = max(abs(cvm - cvg$cvm)),
15   max_diff_cv_se = max(abs(cvsd - cvg$cvstd)),
```

```

16 lambda_min = lgrid[i], glmnet_lambda_min = cvg$lambda.min,
17 lambda_1se = max(lgrid[cvm <= cvm[i] + cvsd[i]]),
18 glmnet_lambda_1se = cvg$lambda.1se)

```

```

## max_diff_cv_curve      max_diff_cv_se      lambda_min
##          2.910e-11          3.638e-12          1.843e+00
## glmnet_lambda_min      lambda_1se glmnet_lambda_1se
##          1.843e+00          6.940e+01          6.940e+01

```

```

1 nz <- function(s) sum(coef(glmnet(X, y, alpha = 1, lambda = lgrid),
2                           s = s)[-1] != 0)
3 cvr <- cv.glmnet(X, y, alpha = 0, foldid = fold)
4 cvo <- mean(sapply(1:10, function(v) {
5   m <- lm(Salary ~ ., data = d[fold != v, ])
6   mean((d$Salary[fold == v] - predict(m, d[fold == v, ]))^2) }))
7 c(cv_lasso = min(cvg$cvm), cv_ridge = min(cvr$cvm), cv_least_squares = cvo,
8   nonzero_at_lambda_min = nz(cvg$lambda.min),
9   nonzero_at_lambda_1se = nz(cvg$lambda.1se))

```

```

##          cv_lasso          cv_ridge
##          113582          113880

```

```
##      cv_least_squares nonzero_at_lambda_min
##              116299              15
## nonzero_at_lambda_1se
##              6
```

Both penalised fits beat least squares, and the one-standard-error rule buys a model with 6 predictors instead of 15 at a cross-validation error that is within one standard error of the best.

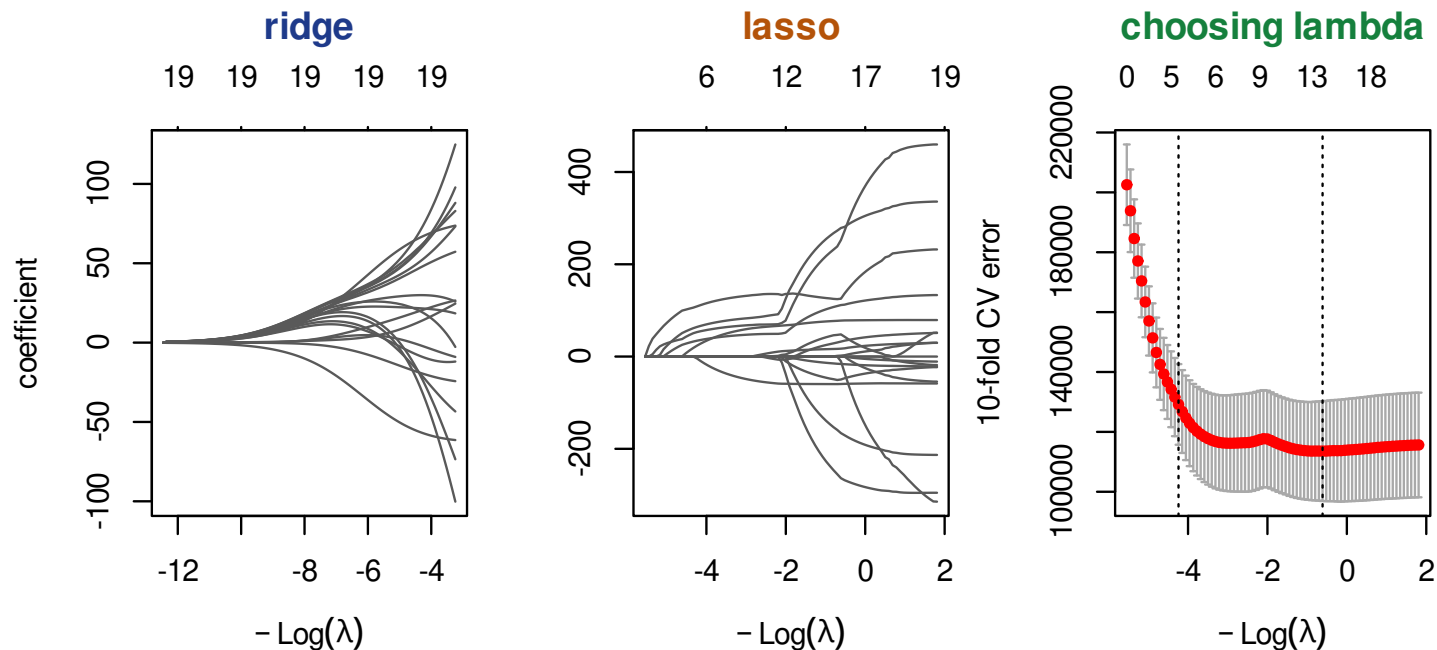


Figure 6.3: Coefficient paths on the standardised **Hitters** design and the cross-validation curve for the lasso. The count along the top of each panel is the number of nonzero coefficients: it stays at 19 for ridge and decreases for the lasso as λ grows, reaching 0 at $\lambda_{\max}$. The dashed lines in the third panel are $\lambda_{\min}$ and λ_{1SE} of (22).

4.6 High dimensions (§3.1, §3.2)

Fifty observations, two hundred predictors, and no relationship whatever between them.

```
1 set.seed(11); nh <- 50; kh <- 200
2 Xh <- matrix(rnorm(nh*kh), nh, kh); yh <- rnorm(nh)
3 mh <- lm(yh ~ Xh)
4 c(rank_of_design = qr(cbind(1, Xh))$rank, n = nh,
5   R2 = summary(mh)$r.squared, SSE = sum(residuals(mh)^2),
6   df_residual = mh$df.residual, NA_coefficients = sum(is.na(coef(mh))))
```

```
##      rank_of_design          n          R2
##              50              50              1
##              SSE      df_residual NA_coefficients
##              0              0              151
```

```
1 ds <- c(10, 30, 45, 49)
2 sub <- sapply(ds, function(dd) {
3   s <- summary(lm(yh ~ Xh[, 1:dd])); c(s$r.squared, s$adj.r.squared) })
4 rbind(predictors = ds, R2 = sub[1, ], adjusted_R2 = sub[2, ])
```

```
##              [,1]      [,2]      [,3] [,4]
## predictors 10.0000 30.0000 45.0000  49
```

```
## R2          0.13312  0.4524  0.9633    1
## adjusted_R2 -0.08915 -0.4124  0.5502  NaN
```

R^2 climbs to 1 on pure noise, as Corollary 6.31 says it must. $\bar{R}^2$, meant to guard against exactly that, is negative at 10 and 30 predictors, rises to 0.55 at 45, and is **NaN** at 49, where the residual degrees of freedom vanish; it orders these models in no useful way. The lasso remains defined, and its active set never exceeds the rank of the centred design, which is Theorem 6.20.

```
1 fh <- glmnet(Xh, yh, alpha = 1, thresh = 1e-14)
2 c(max_active_set = max(fh$df), rank_of_centred_design = nh - 1, k = kh,
3   smallest_lasso_SSE = min(colSums((yh - predict(fh, Xh))^2)))
```

```
##          max_active_set rank_of_centred_design
##          49.00000          49.00000
##          k          smallest_lasso_SSE
##          200.00000          0.03851
```

4.7 Nearest neighbours (§3.3–§3.5)

```
1 knn_manual <- function(xtr, ytr, xte, K) apply(xte, 1, function(x0)
2   mean(ytr[order(sqrt(colSums((t(xtr) - x0)^2)))[1:K])])
3 set.seed(7); ntr <- 200
```

```

4 xtr <- matrix(runif(ntr*2), ntr, 2)
5 ytr <- sin(3*xtr[, 1]) + rnorm(ntr, sd = 0.2)
6 xte <- matrix(runif(50*2), 50, 2)
7 apply(c(1, 5, 20), function(K)
8   max(abs(knn_manual(xtr, ytr, xte, K) - knn.reg(xtr, xte, ytr, k = K)$pred)))

```

```
## [1] 0 0 0
```

Theorem 6.37: the median sup-norm distance from the centre of the unit cube to the nearest of n uniform points, as a function of n and k .

```

1 med <- function(n, k) 0.5 * (1 - 2^(-1/n))^(1/k)
2 tb <- outer(c(100, 500, 5000), c(1, 2, 5, 10, 20), Vectorize(med))
3 dimnames(tb) <- list(paste0("n=", c(100, 500, 5000)),
4   paste0("k=", c(1, 2, 5, 10, 20)))
5 round(tb, 4)

```

```

##           k=1    k=2    k=5    k=10    k=20
## n=100  0.0035 0.0416 0.1849 0.3040 0.3899
## n=500  0.0007 0.0186 0.1341 0.2589 0.3598
## n=5000 0.0001 0.0059 0.0846 0.2057 0.3207

```

```

1 set.seed(2); Bm <- 4000
2 sim <- replicate(Bm,
3   min(apply(abs(matrix(runif(500*10, -0.5, 0.5), 500, 10))), 1, max)))
4 c(simulated_median = median(sim), formula = med(500, 10))

```

```

## simulated_median      formula
##           0.2591          0.2589

```

Proposition 6.38: the side a centred cube needs in order to hold a fraction p of the distribution.

```

1 side <- outer(c(0.01, 0.1, 0.5), c(1, 2, 5, 10, 20), function(p, k) p^(1/k))
2 dimnames(side) <- list(paste0("p=", c(0.01, 0.1, 0.5)),
3   paste0("k=", c(1, 2, 5, 10, 20)))
4 round(side, 3)

```

```

##           k=1    k=2    k=5    k=10   k=20
## p=0.01  0.01  0.100  0.398  0.631  0.794
## p=0.1   0.10  0.316  0.631  0.794  0.891
## p=0.5   0.50  0.707  0.871  0.933  0.966

```

To capture a tenth of the data in ten dimensions the neighbourhood must span 79% of the range of

every coordinate, so it is not local in any useful sense.

Least squares against KNN as the dimension grows, with only the first predictor entering the truth and $\sigma = 0.5$, so the irreducible error is 0.25.

```

1 run <- function(kk, f, ntr = 100, nte = 2000, Ks = c(1, 5, 20), B = 40) {
2   rowMeans(replicate(B, {
3     Xtr <- matrix(runif(ntr*kk, -1, 1), ntr, kk)
4     Xte <- matrix(runif(nte*kk, -1, 1), nte, kk)
5     ytr <- f(Xtr[, 1]) + rnorm(ntr, sd = 0.5)
6     yte <- f(Xte[, 1]) + rnorm(nte, sd = 0.5)
7     o <- lm(ytr ~ Xtr)
8     c(OLS = mean((yte - cbind(1, Xte) %*% coef(o))^2),
9       sapply(Ks, function(K)
10         mean((yte - knn.reg(Xtr, Xte, ytr, k = K)$pred)^2))) })))
11 }
12 set.seed(5); ks <- c(1, 2, 4, 10, 20)
13 lin <- t(sapply(ks, run, f = function(x) 2*x))
14 nlin <- t(sapply(ks, run, f = function(x) sin(3*x) + x^2))
15 colnames(lin) <- colnames(nlin) <- c("OLS", "K=1", "K=5", "K=20")
16 round(cbind(k = ks, lin), 3)

```

```
##          k  OLS   K=1   K=5   K=20
## [1,]    1 0.256 0.511 0.306 0.279
## [2,]    2 0.257 0.521 0.329 0.360
## [3,]    4 0.266 0.714 0.444 0.536
## [4,]   10 0.280 1.227 0.715 0.801
## [5,]   20 0.314 1.709 0.976 1.013
```

```
1 round(cbind(k = ks, nlin), 3)
```

```
##          k  OLS   K=1   K=5   K=20
## [1,]    1 0.514 0.508 0.305 0.295
## [2,]    2 0.518 0.535 0.340 0.370
## [3,]    4 0.529 0.706 0.448 0.506
## [4,]   10 0.563 1.091 0.648 0.643
## [5,]   20 0.653 1.306 0.760 0.720
```

When the truth is linear, least squares wins at every dimension. When it is not, KNN at $K = 5$ wins up to $k = 4$ and loses from $k = 10$ onward. Between $k = 1$ and $k = 20$ its error rises from 0.305 to 0.760 while the least squares error rises only from 0.514 to 0.653: the deterioration is the bias term of (24) responding to Theorem 6.37, and it is not offset by any change in the variance, which by Proposition 6.36 does not involve k at all.

5 Pitfalls

5.1 Penalising unstandardised predictors

By Proposition 6.5 a change of units changes the penalty a predictor carries. Below, one column is re-expressed in thousands, which is no change of information at all.

```

1  Xr <- X; Xr[, "CRuns"] <- Xr[, "CRuns"]/1000
2  g <- function(M, st) min(cv.glmnet(M, y, alpha = 1, foldid = fold,
3                                standardize = st)$cvm)
4  c(standardised_original = g(X, TRUE),  standardised_rescaled = g(Xr, TRUE),
5    raw_original = g(X, FALSE),          raw_rescaled = g(Xr, FALSE))

```

```

## standardised_original standardised_rescaled
##                113582                113582
##      raw_original      raw_rescaled
##                112619                116058

```

Diagnostic. A penalised fit whose answer moves when a column is rescaled. **Fix.** Standardise by (5), inside every cross-validation fold. Note that the unstandardised fit is not uniformly worse here; the objection is that it depends on a choice carrying no information.

5.2 Carrying a tuning constant across parametrisations

λ means different things in (1), in the $(2n)^{-1}$ scaling of `glmnet`, and in the constrained form of Theorem 6.17. Below, the plausible conversion $\lambda = n\lambda_{\text{glmnet}}$ is applied to a ridge fit.

```

1 lg <- 0.38
2 rg <- glmnet(Xs, yc, alpha = 0, lambda = lg*c(4, 2, 1), standardize = FALSE,
3             intercept = FALSE, thresh = 1e-16)
4 b_pkg <- as.matrix(coef(rg))[-1, 3]
5 b_naive <- drop(solve(crossprod(Xs) + n*lg*diag(k), crossprod(Xs, yc)))
6 b_right <- drop(solve(crossprod(Xs) + (n*lg/sdn(y))*diag(k), crossprod(Xs, yc)))
7 c(naive_conversion = max(abs(b_naive - b_pkg)),
8   correct_conversion = max(abs(b_right - b_pkg)),
9   coefficient_scale = max(abs(b_pkg)))

```

```

##      naive_conversion correct_conversion  coefficient_scale
##           379.6218951           0.0008683           424.8559979

```

Diagnostic. Manual and package coefficients disagreeing by an amount comparable to the coefficients themselves, while both satisfy their own normal equations. **Fix.** Recover the implied λ from the stationarity conditions, as in §4.4, rather than trusting a remembered conversion.

5.3 Reading the lasso's active set as the set of important predictors

Theorem 6.27 says the split between correlated predictors is arbitrary. The consequence is visible in the bootstrap: how often does each predictor survive at λ_{1SE} ?

```

1 set.seed(8); Bb <- 200
2 sel <- replicate(Bb, {
3   ii <- sample(n, replace = TRUE)
4   cc <- cv.glmnet(X[ii, ], y[ii], alpha = 1, nfolds = 10)
5   as.numeric(coef(cc, s = "lambda.1se")[-1] != 0) })
6 fr <- rowMeans(sel); names(fr) <- colnames(X)
7 round(sort(fr, decreasing = TRUE), 2)

```

##	Hits	Walks	PutOuts	CRuns	CRBI
##	0.90	0.84	0.80	0.71	0.70
##	DivisionW	CHmRun	CHits	RBI	Runs
##	0.66	0.42	0.32	0.31	0.24
##	LeagueN	Assists	AtBat	Errors	HmRun
##	0.16	0.16	0.14	0.13	0.11
##	CWalks	NewLeagueN	Years	CAtBat	
##	0.10	0.10	0.09	0.00	

```
1 c(mean_active_set_size = mean(colSums(sel)),  
2   sd_active_set_size = sd(colSums(sel)),  
3   selected_in_all_200 = sum(fr == 1))
```

```
## mean_active_set_size    sd_active_set_size  
##                6.930                2.951  
## selected_in_all_200  
##                0.000
```

Diagnostic. Selection frequencies spread across the middle of $[0, 1]$ and an active set whose size varies by several predictors from one resample to the next. **Fix.** Report the selection frequencies rather than one active set, and keep prediction claims separate from claims about which variables matter.

5.4 Quoting the minimum of the cross-validation curve as the test error

λ was chosen to minimise $CV(\lambda)$, so $\min_{\lambda} CV(\lambda)$ is the minimum of a noisy curve and is biased downward as an estimate of the error at the selected λ . An honest figure puts the whole procedure,

the choice of λ included, inside an outer cross-validation.

```

1 nested <- mean(sapply(1:10, function(v) {
2   tr <- fold != v
3   cc <- cv.glmnet(X[tr, ], y[tr], alpha = 1, nfolds = 10)
4   mean((y[!tr] - predict(cc, X[!tr, ], s = "lambda.min"))^2) }))
5 c(minimum_of_cv_curve = min(cvg$cvm), nested_cv = nested,
6   ratio = nested/min(cvg$cvm))

```

## minimum_of_cv_curve	nested_cv	ratio
## 113581.650	120080.905	1.057

Diagnostic. A reported error that came from the same curve used to pick the tuning constant. **Fix.** Nested cross-validation, or a held-out set never seen by the tuning step. This is the pitfall of Unit 1 about cross-validating the whole procedure, with λ in place of the variable selection.

6 Exercises

Exercise 6.1. Show that $\mathbf{X}'\mathbf{X} + \lambda\mathbf{I}$ is invertible for every $\lambda > 0$ whatever the rank of $\mathbf{X}$, and explain what changes at $\lambda = 0$.

Solution. For $\mathbf{c} \neq \mathbf{0}$, $\mathbf{c}'(\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})\mathbf{c} = |\mathbf{X}\mathbf{c}|^2 + \lambda|\mathbf{c}|^2 \geq \lambda|\mathbf{c}|^2 > 0$, so the matrix is positive definite and hence invertible; this is Theorem 6.2. At $\lambda = 0$ the second term disappears and the form vanishes on $\mathcal{N}(\mathbf{X})$, which is nontrivial exactly when $\text{rank}(\mathbf{X}) < k$, for instance whenever $k > n$. Least squares then has the whole affine family of solutions of Proposition 6.32, while ridge has one. ■

Exercise 6.2. An orthonormal design gives least squares coefficients $(2.0, -1.5, 0.8, 0.1)$. Compute the ridge and lasso coefficients at $\lambda = 1$, and state the largest λ at which the lasso fit is still not identically zero.

Solution. By Theorem 6.22, ridge divides by $1 + \lambda = 2$, giving $(1.0, -0.75, 0.4, 0.05)$, and the lasso subtracts $\lambda/2 = 0.5$ from each modulus, giving $(1.5, -1.0, 0.3, 0)$.

With $\mathbf{X}'\mathbf{X} = \mathbf{I}$ the quantities $\mathbf{x}'_j\mathbf{Y}$ are the least squares coefficients, so by Corollary 6.19 the fit is

zero exactly when $\lambda \geq 2 \max_j |\hat{\beta}_j| = 4$. The lasso is therefore not identically zero for every $\lambda < 4$. ■

Exercise 6.3. A standardised design has singular values $d = (4, 2, 1, 0.5)$. Compute $\text{df}(\lambda)$ at $\lambda = 2$ and compare with the number of coefficients estimated. Which direction is shrunk hardest?

Solution. By (10),

$$\text{df}(2) = \frac{16}{18} + \frac{4}{6} + \frac{1}{3} + \frac{0.25}{2.25} = \frac{8}{9} + \frac{2}{3} + \frac{1}{3} + \frac{1}{9} = 2,$$

against $k = 4$ coefficients estimated, none of them zero. The factors are 0.889, 0.667, 0.333, 0.111, so the direction with $d_4 = 0.5$, the one along which the design varies least, is shrunk to a ninth of its least squares value. ■

Exercise 6.4. Prove that $n \text{Err}_{\text{in}} = \mathbf{E}[\text{SSE}] + 2\sigma^2 \text{tr}(\mathbf{S})$ for any linear smoother $\hat{\mathbf{Y}} = \mathbf{S}\mathbf{Y}$, and say where the proof of Unit 5 used idempotency that this one does not.

Solution. The proof is Theorem 6.11. Unit 5 took $\mathbf{S} = \mathbf{H}$ and used $\mathbf{M}'\mathbf{M} = \mathbf{M}$ and $\text{tr}(\mathbf{M}) = n - d$, both consequences of $\mathbf{H}$ being a symmetric idempotent. The general argument keeps $\text{tr}(\mathbf{M}'\mathbf{M})$ and $\text{tr}(\mathbf{S}'\mathbf{S})$ separate and finds that they cancel, leaving $2\sigma^2 \text{tr}(\mathbf{S})$; nothing beyond $\mathbf{S}$ being fixed and $\text{var}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ is used. That is what licenses calling $\text{tr}(\mathbf{S})$ a number of degrees of freedom: it is the quantity that appears in the optimism, whatever the fitting rule. ■

Exercise 6.5. Show that the total mean squared error of the ridge estimator satisfies $R'(0) < 0$ whenever $\mathbf{X}$ has full column rank and $\sigma^2 > 0$, and explain why this does not tell you which λ to use.

Solution. By Theorem 6.14, $R'(\lambda) = \sum_j 2\gamma_j(\lambda\alpha_j^2 - \sigma^2)/(\gamma_j + \lambda)^3$; at $\lambda = 0$ every bracket is $-\sigma^2$ and $R'(0) = -2\sigma^2 \sum_j \gamma_j^{-2} < 0$. So R strictly decreases at 0 and some $\lambda > 0$ has $R(\lambda) < R(0)$. The set on which the decrease is guaranteed is $[0, \sigma^2 / \max_j \alpha_j^2)$, whose right endpoint involves σ^2 and $\boldsymbol{\alpha} = \mathbf{V}'\boldsymbol{\beta}$, both unknown: knowing them would make the estimation problem unnecessary. The theorem is an existence statement, and the choice of λ is made by cross-validation. ■

Exercise 6.6. State and prove the condition for $\hat{\boldsymbol{\beta}}$ to minimise $|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}|^2 + \lambda\|\boldsymbol{\beta}\|_1$. If $\mathbf{x}'_1\mathbf{Y} = 30$, $\mathbf{x}'_2\mathbf{Y} = -45$ and $\mathbf{x}'_3\mathbf{Y} = 12$, at what λ does the fit collapse to zero?

Solution. The condition is (15) and the proof is Theorem 6.18: convexity reduces optimality to the non-negativity of every directional derivative, and taking $\mathbf{v} = \pm\mathbf{e}_j$ decouples the requirement into one condition per coordinate.

By Corollary 6.19 the fit is zero exactly when $\lambda \geq 2 \max_j |\mathbf{x}'_j\mathbf{Y}| = 2 \cdot 45 = 90$. ■

Exercise 6.7. Two columns of the design are identical. What does ridge do with them, what does the elastic net do, and what does the lasso do? Prove each claim.

Solution. All three are Theorem 6.27, whose engine is that exchanging the two coordinates leaves the criterion unchanged. Ridge and the elastic net with $\lambda_2 > 0$ have a unique minimiser by Proposition 6.26, so the minimiser must be its own image under the exchange, forcing the two coefficients to be equal.

The lasso has no such uniqueness: with the two coefficients replaced by $(t, \sigma - t)$ for any t between 0 and their sum σ , the fitted vector is unchanged and the penalty is unchanged, so every point

of that segment is a minimiser. Which one a solver returns is a property of the solver, not of the data. ■

Exercise 6.8. Prove that a solution of $\min |\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}|^2$ subject to $\|\boldsymbol{\beta}\|_1 \leq s$ also solves the penalised problem for some $\lambda \geq 0$. Which hypothesis fails if $s = 0$?

Solution. The proof is Theorem 6.17(ii): the programme is convex, its value is finite, and Slater's condition holds at $\tilde{\boldsymbol{\beta}} = \mathbf{0}$ because $\|\mathbf{0}\|_1 - s = -s < 0$. Strong duality then produces a multiplier $\lambda \geq 0$ at which $\hat{\boldsymbol{\beta}}$ minimises the Lagrangian, which is the penalised criterion up to the constant λs .

If $s = 0$ the Slater point disappears: the only feasible $\boldsymbol{\beta}$ is $\mathbf{0}$ and no strictly feasible point exists. The conclusion is trivially true in that case anyway, since $\boldsymbol{\beta} = \mathbf{0}$ minimises the penalised criterion for $\lambda \geq \lambda_{\max}$. ■

Exercise 6.9. Explain why a lasso fit on a design with $k > n$ can be taken to involve at most n predictors, and why this does not make the selected set trustworthy.

Solution. Theorem 6.20: if the active columns were dependent, one could move along a direction in the null space of $\mathbf{X}$ without changing the fitted vector, and the ℓ_1 penalty is affine along that direction, so it must be constant there; moving to the end of the segment zeroes a coefficient without leaving the solution set. Iterating gives a minimiser with independent active columns, and at most $\text{rank}(\mathbf{X}) \leq n$ columns of $\mathbf{X}$ can be independent.

It does not make the set trustworthy, for the reason in Proposition 6.33(iii): with $k > n$ the columns are dependent, so a predictor in the active set can be exchanged for a combination of others at no cost in fit. The bootstrap of §5.3 shows this happening on data with $k < n$ already.



Exercise 6.10. With $n = 40$ observations and $k = 39$ predictors, what are SSE, R^2 , $\hat{\sigma}^2$, C_p and $\bar{R}^2$? Which of them can be repaired and how?

Solution. $\text{rank}(\mathbf{X}_+) \leq 40 = n$ and generically equals it, so by Theorem 6.30 $\text{SSE} = 0$ and $R^2 = 1$ for any response whatever. Then $\hat{\sigma}^2 = \text{SSE}/(n - k - 1) = 0/0$ is undefined, and C_p , AIC and BIC, which all need $\hat{\sigma}^2$, are undefined with it; $\bar{R}^2 = 1 - \{\text{SSE}/(n - k - 1)\}/\{\text{SST}/(n - 1)\}$ is undefined for the same reason (Corollary 6.31).

None of them can be repaired by an adjustment to the formula, because the failure is not arithmetic: a perfect fit is available for any response, so the training data carries no information about prediction error. What replaces them is an error estimate computed on data not used in the fit, that is cross-validation or a held-out set, together with a fitting rule that is defined at $k \geq n$, such as ridge or the lasso. ■

Exercise 6.11. With $n = 1000$ points uniform on $[-\frac{1}{2}, \frac{1}{2}]^k$, compute the median sup-norm distance from the centre to its nearest neighbour for $k = 1$ and $k = 10$. What side must a centred cube have to contain 5% of the distribution when $k = 20$?

Solution. By (27), $m(n, k) = \frac{1}{2}(1 - 2^{-1/n})^{1/k}$. With $n = 1000$, $1 - 2^{-1/1000} = 6.929 \times 10^{-4}$, so

$$m(1000, 1) = 3.46 \times 10^{-4}, \quad m(1000, 10) = \frac{1}{2}(6.929 \times 10^{-4})^{1/10} = 0.242.$$

The nearest neighbour is 697 times further away in ten dimensions than in one, and at 0.242 it is roughly half the way to the boundary.

By Proposition 6.38 the side is $p^{1/k} = 0.05^{1/20} = 0.861$: capturing a twentieth of the distribution takes 86% of the range of every coordinate. ■

Exercise 6.12. Compare the average variance of the least squares fit and of the K -nearest-neighbour fit at the training inputs. At which K are they equal when $n = 200$ and $k = 9$? Where does the dimension enter the KNN error?

Solution. By Proposition 6.36 the averages are $\sigma^2(k+1)/n$ and σ^2/K , equal when $K = n/(k+1)$, that is $K = 200/10 = 20$.

The dimension does not appear in the KNN variance at all. It enters through the bias term of (24), bounded in Corollary 6.39 by L times the neighbourhood radius, and Theorem 6.37 shows that radius growing quickly with k : the K points being averaged are no longer near x_0 , so their mean is no longer an estimate of $f(x_0)$. ■

Exercise 6.13. In R: (a) reproduce a `glmnet` ridge fit from (2) by recovering the implied λ from the normal equations; (b) verify Theorem 6.11 by simulation for a ridge smoother; (c) show that the lasso active set on `Hitters` varies under resampling.

Solution. (a) is §4.4, (b) §4.2, (c) §5.3. In (a) the point of recovering λ from $\mathbf{x}'_j(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})/\hat{\beta}_j$ is that the ratio must be the same at every j : that is the normal equation, and it certifies the answer

without any assumption about how the package parametrises its penalty. In (b) the simulated difference should match $2\sigma^2 \text{tr}(\mathbf{S})$ to within the printed Monte Carlo standard error, and $\text{tr}(\mathbf{S})$ is not an integer. ■